

PAIR OF STRAIGHT LINES



SYNOPSIS

Homogeneous equations:

Combined Equation of a Pair of Straight lines :

- i) If $L_1 = 0, L_2 = 0$ are any two lines, then the combined equation of $L_1 = 0, L_2 = 0$ is $L_1 L_2 = 0$
- ii) Any second degree equation in x and y represents a pair of straight lines if the expression on the left hand side can be expressed as a product of two linear factors in x and y .

Separate equations of pair of lines :

- The equations of the separate lines of $ax^2 + 2hxy + by^2 = 0$ are

$$ax + \left(h + \sqrt{h^2 - ab}\right)y = 0,$$

$$ax + \left(h - \sqrt{h^2 - ab}\right)y = 0$$

Nature of pair of lines :

- The second degree homogeneous equation $ax^2 + 2hxy + by^2 = 0$ represents a pair of straight lines passing through the origin and it represents
 - (i) two real and distinct lines if $h^2 > ab$
 - (ii) two coincident lines if $h^2 = ab$
 - (iii) Imaginary lines if $h^2 < ab$

W.E-1:- The equation $2x^2 + kxy + 2y^2 = 0$ represents a pair of imaginary lines if

Sol : The given equation will represent a pair of imaginary lines if $h^2 < ab \Rightarrow k^2 < 16$
 $\Rightarrow (k - 4)(k + 4) < 0$
 $\Rightarrow k \in (-4, 4)$

Slopes of pair of lines :

- i) If $y = m_1 x, y = m_2 x$ are the two lines represented by the pair of lines $ax^2 + 2hxy + by^2 = 0, b \neq 0$ with slopes m_1 and m_2 then

- a) The slopes of the lines are the roots of the quadratic equation

$$bm^2 + 2hm + a = 0$$

$$b) m_1 + m_2 = \frac{-2h}{b}; m_1 m_2 = \frac{a}{b}; |m_1 - m_2| = \frac{2\sqrt{h^2 - ab}}{|b|}$$

- c) The combined equation of pair of lines with slopes m_1, m_2 is

$$(y - m_1 x)(y - m_2 x) = 0$$

$$\Rightarrow y^2 - (m_1 + m_2)xy + m_1 m_2 x^2 = 0$$

- ii) The slopes of the straight lines represented by $ax^2 + 2hxy + by^2 = 0$ are reciprocal to each other if $a = b$

- iii) If the slopes of two lines represented by $ax^2 + 2hxy + by^2 = 0$ are in the ratio $l : m$ then

$$(l + m)^2 ab = 4h^2 lm$$

- iv) If the slope of one of the lines represented by $ax^2 + 2hxy + by^2 = 0$ is k times the slope of other line then $4kh^2 = (k + 1)^2 ab$

- v) $ax^2 + 2hxy + by^2 = 0$ represents a pair of lines if the slope of one line is the n^{th} power of the other then

$$(ab^n)^{1/n+1} + (a^n b)^{1/n+1} + 2h = 0$$

- vi) If the slope of one line of pair of lines $ax^2 + 2hxy + by^2 = 0$ is square of the slope of the other line then $ab(a + b - 6h) + 8h^3 = 0$

Angle between the pair of lines :

- If θ is an acute angle between the pair of lines $ax^2 + 2hxy + by^2 = 0$ then

$$\cos \theta = \frac{|a + b|}{\sqrt{(a - b)^2 + 4h^2}} \text{ or}$$

$$\sin \theta = \frac{2\sqrt{h^2 - ab}}{\sqrt{(a - b)^2 + 4h^2}} \text{ or}$$

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{|a+b|} ; a+b \neq 0$$

i) The lines represented by $ax^2 + 2hxy + by^2 = 0$ are perpendicular ,

if $a+b=0$.

i.e., coefficient of x^2 + coefficient of $y^2 = 0$

Pair of parallel & perpendicular lines :

- i) The equation to the pair of lines passing through the point (x_1, y_1) and parallel to the pair of straight lines $ax^2 + 2hxy + by^2 = 0$ is

$$a(x-x_1)^2 + 2h(x-x_1)(y-y_1) + b(y-y_1)^2 = 0$$

ii) The equation to the pair of lines passing through the origin and perpendicular to

$$ax^2 + 2hxy + by^2 = 0 \text{ is}$$

$$bx^2 - 2hxy + ay^2 = 0$$

iii) The equation to the pair of lines passing through the point (x_1, y_1) and perpendicular to the pair of straight lines $ax^2 + 2hxy + by^2 = 0$ is

$$b(x-x_1)^2 - 2h(x-x_1)(y-y_1) + a(y-y_1)^2 = 0$$

Common line to pair of lines :

- i) If the pairs of lines $a_1x^2 + 2h_1xy + b_1y^2 = 0$, $a_2x^2 + 2h_2xy + b_2y^2 = 0$ have one line in common then

$$\begin{vmatrix} a_1 & 2h_1 \\ a_2 & 2h_2 \end{vmatrix} \cdot \begin{vmatrix} 2h_1 & b_1 \\ 2h_2 & b_2 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}^2 \quad (\text{or})$$

$$(a_1b_2 - a_2b_1)^2 + 4(h_1a_2 - h_2a_1)(h_1b_2 - h_2b_1) = 0$$

ii) If one of the lines represented by

$a_1x^2 + 2h_1xy + b_1y^2 = 0$ is perpendicular to one of the lines represented by

$$a_2x^2 + 2h_2xy + b_2y^2 = 0 \text{ then}$$

$$\begin{vmatrix} a_1 & 2h_1 \\ b_2 & -2h_2 \end{vmatrix} \cdot \begin{vmatrix} 2h_1 & b_1 \\ -2h_2 & a_2 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 \\ b_2 & a_2 \end{vmatrix}^2 \quad (\text{or})$$

$$(a_1a_2 - b_1b_2)^2 + 4(h_1a_2 + h_2b_1)(h_1b_2 + h_2a_1) = 0$$

iii) If the pair of lines $a_1x^2 + 2h_1xy + b_1y^2 = 0$ and $a_2x^2 + 2h_2xy + b_2y^2 = 0$ are such that they have one line in common and the remaining lines are perpendicular then

$$h_1 \left(\frac{1}{a_1} - \frac{1}{b_1} \right) = h_2 \left(\frac{1}{a_2} - \frac{1}{b_2} \right)$$

Types of triangles :

- i) The equation of the pair of lines passing through the origin and forming an isosceles triangle with the line $ax + by + c = 0$ is

$$(ax + by)^2 - k(bx - ay)^2 = 0.$$

(a) If $k = 1$ then the triangle is right angled isosceles.

(b) If $k = 3$ then the triangle is equilateral.

(c) If $k = \frac{1}{3}$ then the triangle is an isosceles and obtuse angled

ii) The triangle formed by the pair of lines $S \equiv ax^2 + 2hxy + by^2 = 0$ and the line

$$lx + my + n = 0 \text{ is}$$

a) equilateral if $ax^2 + 2hxy + by^2 =$

$$(lx + my)^2 - 3(mx - ly)^2 =$$

b) Isosceles if $h(l^2 - m^2) = (a - b)lm$

c) Right angled if $a + b = 0$ or $S(l, m) = 0$

W.E-2:- The triangle formed by the

lines $2x^2 + 3xy - 2y^2 = 0, 3x + y + 1 = 0$ is

Sol: Given line is $3x + y + 1 = 0$

The simplification of $(3x+y)^2 - (x-3y)^2 = 0$ is given pair of lines

It is in the form $(ax + by)^2 - k(bx - ay)^2 = 0$

$$\text{Here } k = 1$$

∴ The triangle is right angled isosceles.

Centres related with triangles :

- i) If (α, β) is the centroid of the triangle whose sides are $ax^2 + 2hxy + by^2 = 0$ and $lx + my + n = 0$, then

$$\frac{\alpha}{bl-hm} = \frac{\beta}{am-hl} = \frac{-2n}{3(bl^2-2hlm+am^2)} \text{ (or)}$$

$$\frac{\alpha}{\left(\frac{\partial F}{\partial x}\right)_{(l,m)}} = \frac{\beta}{\left(\frac{\partial F}{\partial y}\right)_{(l,m)}} = \frac{-n}{3F_{(l,m)}}$$

where $F = bx^2 - 2hxy + ay^2$

ii) The pair of lines $S \equiv ax^2 + 2hxy + by^2 = 0$

represents two sides of a triangle and (x_1, y_1) is the mid point of the third side then the equation of third side is $S_1 = S_{11}$ i.e.,

$$axx_1 + h(xy_1 + x_1y) + byy_1 = ax_1^2 + 2hx_1y_1 + by_1^2$$

iii) If $ax^2 + 2hxy + by^2 = 0$ represents two sides of a triangle, $G(x_1, y_1)$ be its centroid then the mid point of the third side of the triangle is

$$\frac{3}{2}G \text{ i.e., } \left(\frac{3x_1}{2}, \frac{3y_1}{2}\right)$$

iv) If (kl, km) is the orthocentre of the triangle formed by the lines $ax^2 + 2hxy + by^2 = 0$

and $lx + my + n = 0$ then $k = \frac{-n(a+b)}{am^2 - 2hlm + bl^2}$

v) The distance from the origin to the orthocentre of the triangle formed by the lines $\frac{x}{\alpha} + \frac{y}{\beta} = 1$ and $ax^2 + 2hxy + by^2 = 0$ is

$$\sqrt{\alpha^2 + \beta^2} \left| \frac{(a+b)\alpha\beta}{a\alpha^2 - 2h\alpha\beta + b\beta^2} \right|$$

vi) If $ax^2 + 2hxy + by^2 = 0$ represents two sides of a triangle for which (c, d) is the orthocentre, then the equation of the third side of triangle is $(a+b)(cx + dy) = ad^2 - 2hcd + bc^2$

Product of perpendiculars :

- i) The product of the perpendiculars from (α, β) to the pair of lines

$$ax^2 + 2hxy + by^2 = 0 \text{ is } \frac{|a\alpha^2 + 2h\alpha\beta + b\beta^2|}{\sqrt{(a-b)^2 + 4h^2}}$$

Area of the triangle :

- i) The area of the triangle formed by the line $lx + my + n = 0$ and the pair of lines

$$ax^2 + 2hxy + by^2 = 0 \text{ is } \frac{n^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$$

ii) The equation of the pair of lines through the origin and making an angle ' α ' with the line $lx + my + n = 0$ is

$$(lx + my)^2 - \tan^2 \alpha (mx - ly)^2 = 0 \text{ and}$$

$$\text{the area of the triangle is } \frac{n^2}{\tan \alpha (l^2 + m^2)}$$

iii) The area of an equilateral triangle formed by the line $ax + by + c = 0$ with the pair of lines

$$(ax + by)^2 - 3(bx - ay)^2 = 0 \text{ is } \frac{c^2}{\sqrt{3}(a^2 + b^2)}$$

$= \frac{p^2}{\sqrt{3}}$ where p is the perpendicular distance from the origin to the line $ax + by + c = 0$

W.E-3:- If $(\sqrt{2}x + \sqrt{3}y)^2 - 36(\sqrt{3}x - \sqrt{2}y)^2 = 0$

and $\sqrt{2}x + \sqrt{3}y + 4\sqrt{5} = 0$ represents an isosceles triangle with base angle $\tan^{-1} 6$ then its area is

Sol : Equation of given line is $\sqrt{2}x + \sqrt{3}y + 4\sqrt{5} = 0$

here $l = \sqrt{2}$, $m = \sqrt{3}$, $n = 4\sqrt{5}$

Given that $\tan \alpha = 6$

$$\begin{aligned} \therefore \text{Area of the triangle} &= \frac{n^2}{\tan \alpha (l^2 + m^2)} \\ &= \frac{16 \times 5}{6 \times 5} = \frac{8}{3} \text{ sq. units} \end{aligned}$$

Pair of angular bisectors :

- i) The equation to the pair of bisectors of the angles between the pair of straight lines

$$ax^2 + 2hxy + by^2 = 0 \text{ is } h(x^2 - y^2) = (a - b)xy$$

ii) The angle between pair of angular bisectors of any pair of lines is $\frac{\pi}{2}$.

iii) The equation to the pair of bisectors of the co-ordinate axes is $x^2 - y^2 = 0$

iv) If one of the line in $ax^2 + 2hxy + by^2 = 0$ bisects the angle between the coordinate axes then $(a + b)^2 = 4h^2$

W.E-4:- Equation of the bisectors of the angles between the lines through the origin and the sum and product of whose slopes are respectively the arithmetic and geometric means of 9 and 16 is

Sol: A.M. of 9 and 16 = $\frac{9+16}{2} = \frac{25}{2} = m_1 + m_2$

G.M. of 9 and 16 = $\sqrt{9 \times 16} = 12 = m_1 \cdot m_2$

Equation of pair of lines is

$$y^2 - (m_1 + m_2)xy + m_1m_2x^2 = 0$$

$$\Rightarrow 24x^2 - 25xy + 2y^2 = 0$$

Equation of the bisectors is

$$h(x^2 - y^2) = (a - b)xy$$

$$\Rightarrow 25x^2 + 44xy - 25y^2 = 0$$

Equally inclined with a line :

- i) A pair of lines $L_1L_2 = 0$ is said to be equally inclined to a line $L = 0$ if the lines $L_1 = 0, L_2 = 0$ subtend the same angle with the line $L = 0$
- ii) Every pair of lines is equally inclined to either of its angular bisectors
- iii) A pair of lines is equally inclined to a line $L = 0$, if $L = 0$ is parallel to one of the angular bisectors.
- iv) Given pair of lines through origin is equally inclined to the coordinate axes $\Leftrightarrow$ the pair of angular bisectors of given pair of lines through origin is the coordinate axes
- v) If the pair of lines $ax^2 + 2hxy + by^2 = 0$ equally inclined to the coordinate axes then $h = 0$ and

$$ab < 0$$

vi) The pair of lines $L_1L_2 = 0$ bisects the angle between the pair of lines $L_3L_4 = 0 \Leftrightarrow$ pair of angular bisectors of $L_3L_4 = 0$ and pair of lines $L_1L_2 = 0$ represents the same equation

vii) Two pairs of lines $L_1L_2 = 0, L_3L_4 = 0$ are such that each bisects the angle between the other pair $\Leftrightarrow$ pair of angular bisector of $L_1L_2 = 0$, pair of lines $L_3L_4 = 0$ represents same and vice versa.

viii) Two pairs of lines are equally inclined to each other $\Leftrightarrow$ two pairs of lines have same pair of angular bisectors

W.E-5:- The lines $ax^2 + 2hxy + by^2 = 0$ are equally inclined to the lines

$ax^2 + 2hxy + by^2 + \lambda(x^2 + y^2) = 0$ for what values of λ ?

Sol: Equation of the bisectors of the angle between the lines $ax^2 + 2hxy + by^2 + \lambda(x^2 + y^2) = 0$ is $h(x^2 - y^2) = (a - b)xy$. Which is same as the equation of the bisectors of angles between the lines $ax^2 + 2hxy + by^2 = 0$
 $\therefore$ The given two pairs of lines are equally inclined to each other for any value of λ .

Non homogeneous equations:

Condition for pair of lines :

➤ i) If the equation

$$S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

represents a pair of lines then

$$a) \Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\text{i.e. } \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$$b) h^2 \geq ab, g^2 \geq ac, f^2 \geq bc$$

W.E-6:- If $ax^2 + by^2 + 2fy + c = 0, (a \neq 0)$ represents a pair of lines then f is

Sol : $ax^2 + by^2 + 2fy + c = 0, (a \neq 0)$

represents a pair of lines
then it satisfy the condition $\Delta = 0$

$$\text{i.e. } abc - af^2 = 0 \Rightarrow f^2 = bc$$

$\therefore f$ is G.M. between b and c .

ii) If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of lines then

$ax^2 + 2hxy + by^2 = 0$ represents a pair of lines parallel to them and passing through the origin

Angle between the pair of lines :

➤ i) The angle between the pair of lines

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is same as the angle between the pair of lines $ax^2 + 2hxy + by^2 = 0$

Distance between the pair of lines :

➤ The equation

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of parallel lines iff

$$\Delta = 0, f^2 \geq bc, g^2 \geq ac, h^2 = ab \text{ and } af^2 = bg^2 \quad \text{or}$$

$$\frac{a}{h} = \frac{h}{b} = \frac{g}{f} \text{ and the distance between the parallel}$$

$$\text{lines is } 2\sqrt{\frac{g^2 - ac}{a(a+b)}} \text{ or } 2\sqrt{\frac{f^2 - bc}{b(a+b)}}$$

W.E-7:- The equation

$9x^2 + 2hxy + 4y^2 + 6x + 2fy + c = 0$ represents a pair of parallel straight lines then the relation between h, f and c ($h, f > 0$)

Sol : The given equation represents a pair of parallel

lines then $h^2 = ab, f^2 \geq bc, g^2 \geq ac$

$$\text{and } af^2 = bg^2$$

we get $h = 6, f = 2$ and $c \leq 1$

$\therefore$ relation is $c < f < h$

Product of perpendiculars :

➤ i) The product of the perpendiculars drawn from (α, β) to the pair of lines

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ is}$$

$$\frac{|a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c|}{\sqrt{(a-b)^2 + 4h^2}}$$

ii) The product of the perpendiculars from origin to the pair of lines

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ is}$$

$$\frac{|c|}{\sqrt{(a-b)^2 + 4h^2}}$$

iii) If the pair of lines

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ are}$$

equidistant from the origin then

$$f^4 - g^4 = c(bf^2 - ag^2)$$

Point of intersection of pair of lines:

➤ i) If $S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

represents a pair of lines and $h^2 > ab$ then the point of intersection of the lines is

$$\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right) \text{ i.e., obtained by solving}$$

$$\frac{\partial S}{\partial x} = 0 \text{ and } \frac{\partial S}{\partial y} = 0$$

ii) If the pair of lines

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ intersect at}$$

(α, β) then (α, β) satisfy the equations

$$ax + hy + g = 0, hx + by + f = 0 \text{ and } gx + fy + c = 0$$

$$\text{i.e., } (\alpha, \beta) = \left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$$

$$= \left(\frac{bc - f^2}{hf - bg}, \frac{fg - ch}{hf - bg} \right) = \left(\frac{hc - gf}{af - gh}, \frac{g^2 - ac}{af - gh} \right)$$

iii) The coordinates of the point of intersection of the lines represented by $S = 0$ is

$$\left(\sqrt{\frac{f^2 - bc}{h^2 - ab}}, \sqrt{\frac{g^2 - ac}{h^2 - ab}} \right)$$

iv) If the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ represents}$$

a pair of intersecting lines, then the square of the distance of their point of intersection from

$$\text{the origin is } \frac{c(a+b) - f^2 - g^2}{ab - h^2}$$

v) If the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

represents a pair of perpendicular lines, then the square of the distance of their point of intersection

from the origin is $\frac{f^2 + g^2}{h^2 - ab}$ (or) $\frac{f^2 + g^2}{a^2 + h^2}$ (or)

$$\frac{f^2 + g^2}{b^2 + h^2}$$

Area of the triangle :

- The area of the triangle formed by the line $lx + my + n = 0$ and the pair of lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ whose point of intersection is (x_1, y_1) is

$$\frac{(lx_1 + my_1 + n)^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|} \text{ sq. units}$$

W.E-8:- The area of the triangle whose combined equation

$$(2x^2 - xy - y^2 + x + 2y - 1)(x + y + 1) = 0$$

is

Sol : Given lines are $2x^2 - xy - y^2 + x + 2y - 1 = 0$
 $\rightarrow (1)$

$$\text{and } x + y + 1 = 0 \rightarrow (2)$$

$$\text{From (1) } a = 2, h = -1/2, b = -1$$

$$\text{From (2) } l = 1, m = 1, n = 1$$

Point of intersection of given pair of lines

$$= \left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right) = (0, 1) = (x_1, y_1)$$

Area of the triangle =

$$\frac{(lx_1 + my_1 + n)^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|} = \frac{4 \times 3}{4} = 3 \text{ sq. units}$$

Intercepts of a pair of lines on coordinate axes :

- i) Length of the intercept made by the pair of lines represented by

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ on}$$

$$\text{a) } x\text{-axis is } \frac{2\sqrt{g^2 - bc}}{|a|}$$

$$\text{b) } y\text{-axis is } \frac{2\sqrt{f^2 - bc}}{|b|}$$

ii) If the pair of lines

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

intersect on

$$\text{a) } x\text{-axis, then } g^2 = ac \text{ and}$$

$$2fgh = af^2 + ch^2$$

$$\text{b) } y\text{-axis, then } f^2 = bc \text{ and}$$

$$2fgh = bg^2 + ch^2$$

Pair of angular bisectors :

- If (α, β) be the point of intersection of the pair of lines

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then the equation to the pair of angular bisectors is

$$h[(x - \alpha)^2 - (y - \beta)^2] = (a - b)(x - \alpha)(y - \beta)$$

W.E-9:-

$$\text{If } a(x - 1)^2 + 2h(x - 1)(y - 2) + b(y - 2)^2 = 0$$

has one angular bisector $2x + 3y - 8 = 0$ then other bisector is

Sol : Equation of given angular bisector is

$$2x + 3y - 8 = 0$$

Equation of required angular bisector is

$$3x - 2y + k = 0$$

It passes through the point (1, 2)

$$\therefore \text{required equation is } 3x - 2y + 1 = 0$$

Quadrilateral formed by $S = 0$ and $S^1 = 0$:

- i) The pair of lines $S \equiv ax^2 + 2hxy + by^2 = 0$ and $S^1 \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ forms a

a) rhombus

$$\Leftrightarrow a + b \neq 0, (a - b)fg + h(f^2 - g^2) = 0$$

b) square

$$\Leftrightarrow a + b = 0, (a - b)fg + h(f^2 - g^2) = 0$$

c) rectangle

$$\Leftrightarrow a + b = 0, (a - b)fg + h(f^2 - g^2) \neq 0$$

d) parallelogram

$$\Leftrightarrow a + b \neq 0, (a-b)fg + h(f^2 - g^2) \neq 0$$

e) Area of the parallelogram is $\frac{|c|}{2\sqrt{h^2 - ab}}$

f) Equation of diagonal not passing through origin is $2gx + 2fy + c = 0$ i.e., $S^1 - S = 0$

g) Equation of diagonal passing through origin is $(hf - bg)y = (gh - af)x$

ii) If $ax^2 + 2hxy + by^2 = 0$ are two sides of a parallelogram and $lx + my + n = 0$ is one of the diagonals of the parallelogram then the equation of other diagonal is $(bl - hm)y = (am - hl)x$

iii) Given (x_1, y_1) as opposite vertex of a parallelogram with $S \equiv ax^2 + 2hxy + by^2 = 0$ as one pair of sides then the equation of the diagonal not passing through the origin is $2S_1 - S_{11} = 0$

iv) The pair of lines $xy + ax + by + ab = 0$, $xy + cx + dy + cd = 0$ form a square

a) If $|a - c| = |b - d|$

b) area is $|a - c||b - d|$

c) point of intersection of diagonals is $\left(\frac{-(b+d)}{2}, \frac{-(a+c)}{2}\right)$

d) Equation of diagonals are

$$(a - c)x + (b - d)y + ab - cd = 0$$

$$(a - c)x - (b - d)y + ad - bc = 0$$

W.E-10:- The equation

$x^2y^2 - 9y^2 + 6x^2y + 54y = 0$ represents

Sol: $x^2y^2 - 9y^2 + 6x^2y + 54y = 0$

$$\Rightarrow y^2(x^2 - 9) - 6y(x^2 - 9) = 0$$

$$\Rightarrow y(y - 6)(x - 3)(x + 3) = 0$$

$$\Rightarrow y = 0, y = 6, x = 3, x = -3$$

$\therefore$ The given equation represents four straight lines which form a square.

Homogenisation :

- i) The Combined equation to the pair of lines joining the origin to the points of intersection of the curve $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ and the line $lx + my + n = 0$ is $ax^2 + 2hxy + by^2 +$

$$(2gx + 2fy)\left(\frac{lx + my}{-n}\right) + c\left(\frac{lx + my}{-n}\right)^2 = 0$$

ii) The condition that the pair of lines joining the origin to the points of intersection of

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ and}$$

$$lx + my + n = 0 \text{ to be perpendicular is}$$

$$n^2(a + b) - 2n(lg + mf) + c(l^2 + m^2) = 0$$

W.E-11:- All chords of the curve

$3x^2 - y^2 - 2x + 4y = 0$ which subtend a right angle at the origin, pass through the fixed point

Sol : Given curve is $3x^2 - y^2 - 2x + 4y = 0 \rightarrow (1)$

Let $lx + my = 1 \rightarrow (2)$ be the chord

Homogenising (1) with the help of (2)

$$\text{we get } 3x^2 - y^2 - (2x - 4y)(lx + my) = 0$$

$$\therefore \text{coefficient of } x^2 + \text{coefficient of } y^2 = 0$$

$$\Rightarrow l - 2m = 1 \rightarrow (3)$$

from (2) and (3), we get $(x, y) = (1, -2)$

Some standard results :

- i) The equation to the pair of lines passing through the origin and each is at a distance of d from (α, β)

$$\text{is } (\beta x - \alpha y)^2 = d^2(x^2 + y^2).$$

ii) If L, M are the feet of the perpendiculars from $(c, 0)$ to the lines $ax^2 + 2hxy + by^2 = 0$ then the angle made by LM with positive X-axis is

$$\tan^{-1}\left(\frac{b-a}{2h}\right) \text{ and the equation of LM is}$$

$$(b-a)x - 2hy - bc = 0$$

iii) Point of intersection of diagonals of a rect angle formed by the pairs

$$a_1x^2 + b_1x + c_1 = 0,$$

$$a_2y^2 + b_2y + c_2 = 0 \text{ is } \left(\frac{-b_1}{2a_1}, \frac{-b_2}{2a_2}\right)$$

iv) The image of pair of lines $f(x, y) = 0$ with re

spect to x-axis is $f(x, -y) = 0$ and with respect to y-axis is $f(-x, y) = 0$



- The equation $ax^2 + 2\sqrt{ab}xy + by^2 = 0$ represents a pair of lines which are**
 1) perpendicular 2) coincident
 3) imaginary 4) parallel
- If one of the lines represented by $ax^2 + 2hxy + by^2 = 0$ is the Y-axis then the equation of the other line is**
 1) $ax + 2hy = 0$ 2) $2hx + by = 0$
 3) $ax + by = 0$ 4) $hx + by = 0$
- $y = mx$ is one of the lines represented by $ax^2 + 2hxy + by^2 = 0$ if $a + 2hm + bm^2 =$**
 1) -1 2) 1 3) 0 4) 2
- If the pair of straight lines $Ax^2 + 2Hxy + By^2 = 0$ ($H^2 > AB$) forms an equilateral triangle with the line $ax + by + c = 0$ then $(A + 3B)(3A + B) =$**
 1) H^2 2) $-H^2$ 3) $2H^2$ 4) $4H^2$
- The equation of the line common to the pair of lines $(p^2 - q^2)x^2 + (q^2 - r^2)xy + (r^2 - p^2)y^2 = 0$ and $(l - m)x^2 + (m - n)xy + (n - l)y^2 = 0$ is**
 1) $x + y = 0$ 2) $x - y = 0$
 3) $x + y = pqr$ 4) $x - y = pqr$
- The equation of the line common to the pair of lines $m^2x^2 - (m^2 + 1)xy + y^2 = 0$ and $mx^2 - (m + 1)xy + y^2 = 0$ is**
 1) $mx - y = 0$ 2) $x + y = 0$
 3) $x - y = 0$ 4) $x + y = m$
- If $x^2 + axy + by^2 = 0$ and $x^2 + bxy + cay^2 = 0$ have a common line then the equation of that line is**
 1) $y = cx$ 2) $x = cy$ 3) $x = ay$ 4) $x = by$
- The locus of the points equidistant from the pair of lines $x^2 \cos^2 \theta - 2xy \sin^2 \theta - y^2 \sin^2 \theta = 0$ is**
 1) $x^2 - y^2 + xy \cos ec^2 \theta = 0$
 2) $x^2 + y^2 + 2xy \cos ec^2 \theta = 0$
 3) $x^2 - y^2 + 2xy \sec^2 \theta = 0$
 4) $x^2 + y^2 + 2xy \sec^2 \theta = 0$

- The separate equations of the angular bisectors of the pair of lines $(ax + by)^2 - (bx - ay)^2 = 0$ are**
 1) $ax - by = 0, bx - ay = 0$ 2) $ax + by = 0, bx - ay = 0$
 3) $ax + by = 0, bx + ay = 0$ 4) $(x + y) = 0, (x - y) = 0$

- The slope of angular bisectors of pair of lines $(ax + by)^2 = c(bx - ay)^2, (c > 0)$ are**
 1) $-\frac{a}{b}, \frac{b}{a}$ 2) $\frac{a}{b}, -\frac{b}{a}$ 3) $\frac{a}{c}, -\frac{c}{a}$ 4) $\frac{c}{a}, -\frac{a}{c}$

- The equation to the image of the pair of lines $ax^2 + 2hxy + by^2 = 0$ where $h^2 > ab$ with respect to $y = 0$ is**
 1) $bx^2 + 2hxy + ay^2 = 0$
 2) $bx^2 - 2hxy + ay^2 = 0$
 3) $ax^2 - 2hxy + by^2 = 0$
 4) $ax^2 + 2hxy + by^2 = 0$

- Point of intersection of pair of lines $a(x - \alpha)^2 + 2h(x - \alpha)(y - \beta) + b(y - \beta)^2 = 0$ is**
 1) $(0, 0)$ 2) (α, β)
 3) $(-\alpha, -\beta)$ 4) $(-\alpha, \beta)$

- If the pair of lines given by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ intersect on X-axis then**
 1) b, f, c are in A.P 2) a, f, c are in G.P
 3) a, g, c are in G.P 4) a, g, c are in A.P

- The product of the perpendiculars from $(f, -g)$ to the pair of st.lines $hxy + gx + fy + c = 0$ is**
 1) $|fg|$ 2) $\left| \frac{fg - c}{h} \right|$
 3) $|fgh - c|$ 4) $\left| \frac{fgh - c}{h} \right|$

- The rectangle formed by the pair of lines $2hxy + 2gx + 2fy + c = 0$ with the coordinate axes has the area equal to**
 1) $\frac{|fg|}{h^2}$ 2) $\frac{|gh|}{f^2}$ 3) $\frac{|hf|}{g^2}$ 4) $\frac{|fg|}{h}$

16. If the pair of lines $2hxy+2gx+2fy+c=0$ and the coordinate axes form a rectangle, then the equations of its diagonals are

- 1) $2gx+2fy+c=0, gx-fy=0$
- 2) $2gx+2fy-c=0, gx+fy=0$
- 3) $gx+fy-c=0, gx-fy=0$
- 4) $gx+fy=0, gx-fy=0$

17. The lines $ax^2+2hxy+by^2+2gx+2fy+c=0$ intersect x-axis in A, B and y-axis in C, D respectively. Then the combined equation of $\overline{AB}$ and $\overline{CD}$ is

- 1) $xy=0$
- 2) $ax^2+2hxy+by^2+2gx+2fy+c=0$
- 3) $ax^2+2hxy+by^2+2gx+2fy+c+\frac{4gf}{c}xy=0$
- 4) $ax^2-2hxy+by^2+2gx+2fy+c+\frac{4gf}{c}xy=0$

18. The locus represented by the equation

$$(x-y+c)^2+(x+y-c)^2=0 \text{ is}$$

- 1) A line parallel to x-axis
- 2) A point
- 3) Pair of lines
- 4) Line parallel to y-axis

C.U.Q - KEY

- | | | | | |
|-------|-------|-------|-------|-------|
| 1) 2 | 2) 1 | 3) 3 | 4) 4 | 5) 2 |
| 6) 3 | 7) 2 | 8) 1 | 9) 2 | 10) 1 |
| 11) 3 | 12) 2 | 13) 3 | 14) 4 | 15) 1 |
| 16) 1 | 17) 1 | 18) 2 | | |

C.U.Q - HINTS

1. $h^2 = (\sqrt{ab})^2 = ab$
2. $ax^2+2hxy+by^2 = x(lx+my)$. Coefficient of $x^2 \Rightarrow a=l$, coefficient of $xy \Rightarrow 2h=m$
3. Put $y=mx$ in pair of lines
4. $\cos 60^\circ = \frac{|A+B|}{\sqrt{(A-B)^2+4H^2}}$
5. In both pair of lines sum of the coefficients is zero. The common line is $y=x$,
6. In both pair of lines sum of the coefficients is zero. The common line is $y=x$,

7. $S-S^1=0$

8. Locus is pair of angular bisectors between the pair of lines

9. Given pair is

$$(a^2-b^2)x^2+4abxy+(b^2-a^2)y^2=0.$$

Write the equation of angular bisectors

10. Bisectors are parallel and perpendicular to the line $ax+by+k=0$

11. $f(x, -y) = 0$

12. lines passing through (α, β)

13. $g^2 = ac$

14.
$$\frac{|a\alpha^2+2h\alpha\beta+b\beta^2+2g\alpha+2f\beta+c|}{\sqrt{(a-b)^2+4h^2}}$$

15. $\Delta = 0$ and apply $\frac{|c|}{2\sqrt{h^2-ab}}$

16. Diagonals are

$$2gx+2fy+c=0, (gh-af)x=(hf-bg)y$$

17. Since the given pair intersects x-axis at A, B then the equation of AB is $y=0$, Since the pair cuts y-axis at C, D then the equation of CD is $x=0$. Then the combined equation of $\overline{AB}$ and $\overline{CD}$ is $xy=0$

18. $x-y+c=0, x+y-c=0$ solving we get $(0, c)$ which is a point



LEVEL - I (C.W)

Homogeneous pair of lines

separate equations of pair of lines:

1. The separate equations of the lines

$$6x^2+5xy-6y^2=0 \text{ are}$$

- 1) $2x-3y=0; 3x+2y=0$
- 2) $2x+3y=0; 3x-2y=0$
- 3) $2x-3y=0; 3x-2y=0$
- 4) $3x+2y=0; 2x+3y=0$

Nature of pair of lines:

2. The range of 'a' so that

$$a^2x^2+2xy+4y^2=0 \text{ represents distinct lines}$$

- 1) $a > \frac{1}{2}$ or $a < -\frac{1}{2}$
- 2) $\frac{-1}{2} \leq a \leq \frac{1}{2}$
- 3) $\frac{-1}{2} < a < \frac{1}{2}$
- 4) $a \geq \frac{1}{2}$ or $a \leq -\frac{1}{2}$

Slopes of lines of pair of lines:

3. The difference of the slopes of the lines

$$3x^2 - 4xy + y^2 = 0 \text{ is}$$

- 1) 1 2) 2 3) 3 4) 4

4. The difference of the slopes of the lines represented by

$$x^2(\sec^2 \theta - \sin^2 \theta) - (2 \tan \theta)xy + y^2 \sin^2 \theta = 0$$

- 1) 1 2) 2 3) 3 4) 4

5. The combined equation to a pair of straight lines passing through the origin and inclined at an angles 30° and 60° respectively with X-axis is

1) $\sqrt{3}(x^2 + y^2) = 4xy$

2) $4(x^2 + y^2) = \sqrt{3}xy$

3) $x^2 + \sqrt{3}y^2 - 2xy = 0$

4) $x^2 + 3y^2 - 2xy = 0$

6. If the slope of one line is twice the slope of the other in the pair of straight lines

$$ax^2 + 2hxy + by^2 = 0 \text{ then } 8h^2 =$$

- 1) 7 ab 2) -7ab 3) 9 ab 4) -9ab

7. The equation of the pair of lines passing through the origin whose sum and product of slopes are respectively the arithmetic mean and geometric mean of 4 and 9 is

(EAM-2013)

1) $12x^2 - 13xy + 2y^2 = 0$

2) $12x^2 + 13xy + 2y^2 = 0$

3) $12x^2 - 15xy + 2y^2 = 0$

4) $12x^2 + 15xy - 2y^2 = 0$

8. **Assertion A :** If $ax^2 + 2hxy + by^2 = 0$ represents two straight lines whose one slope is thrice the other then $3h^2 = 4ab$

Reason R: If $ax^2 + 2hxy + by^2 = 0$ represents two lines and whose slopes are m:n then

$$\frac{(m+n)^2}{4mn} = \frac{ab}{h^2}$$

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true and R is not correct explanation of A

3) A is true but R is false

4) A is false but R is true

9. If the sum of the slopes of the lines given by $x^2 + 2cxy + y^2 = 0$ is eight times their product, then c has the value

- 1) 1 2) -1 3) -4 4) -2

Angle between the pair of lines:

10. Angle between the lines

$$x^2 + 2xy \sec \alpha + y^2 = 0 \text{ is}$$

- 1) $\frac{\pi}{2}$ 2) α 3) 2α 4) $\frac{\alpha}{2}$

11. The angle between the pair of lines

$$y^2 \cos^2 \theta - xy \cos^2 \theta + x^2 (\sin^2 \theta - 1) = 0 \text{ is}$$

- 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{6}$ 4) $\frac{\pi}{2}$

12. If the pair of lines given by

$$(x^2 + y^2) \sin^2 \alpha = (x \cos \alpha - y \sin \alpha)^2 \text{ are perpendicular to each other then } \alpha =$$

(EAM-2012)

- 1) $\pi/2$ 2) 0 3) $\pi/4$ 4) $\pi/3$

13. If θ is the acute angle between the pair of lines $x^2 + 3xy - 4y^2 = 0$ then $\sin \theta =$

- 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{3}$ 3) $\frac{5}{\sqrt{34}}$ 4) $\frac{3}{\sqrt{34}}$

Pair of parallel and perpendicular lines:

14. The equation to the pair of lines passing through the point $(-2, 3)$ and parallel to the pair of lines $x^2 + 4xy + y^2 = 0$ is

1) $x^2 - 4xy + y^2 - 8x + 2y - 11 = 0$

2) $x^2 + 4xy + y^2 - 8x + 2y - 11 = 0$

3) $x^2 + 4xy - y^2 - 8x + 2y - 11 = 0$

4) $x^2 - 4xy + y^2 - 8x - 2y - 11 = 0$

15. The equation to the pair of lines passing through the origin and perpendicular to

$$5x^2 + 3xy = 0$$

- 1) $5xy + 3y^2 = 0$ 2) $x^2 - 2y^2 = 0$
3) $3xy - 5y^2 = 0$ 4) $3x^2 - 2xy = 0$

Product of perpendiculars:

16. The product of the perpendiculars from $(-1, 2)$ to the pair of lines

$$2x^2 - 5xy + 2y^2 = 0$$

- 1) 4 2) 3 3) 8 4) $5/2$

17. If the product of perpendiculars from

$$(k, k) \text{ to the pair of lines } x^2 + 4xy + 3y^2 = 0$$

is $4/\sqrt{5}$ then k is

- 1) ± 4 2) ± 3 3) ± 2 4) ± 1

Area of the triangle:

18. Area of the triangle formed by the lines

$$2x - y = 6 \text{ and } 3x^2 - 4xy + y^2 = 0 \text{ is}$$

- 1) 16 2) 25 3) 36 4) 49

19. If the area of the triangle formed by the pair of lines $8x^2 - 6xy + y^2 = 0$ and the line $2x + 3y = a$ is 7 then a = (EAM-2012)

- 1) 14 2) $14\sqrt{2}$ 3) $28\sqrt{2}$ 4) 28

20. If the area of the triangle formed by the lines $3x^2 - 2xy - 8y^2 = 0$ and the line $2x + y - k = 0$ is 5 sq. units, then k =

- 1) 5 2) 6 3) 7 4) 8

21. The area of the equilateral triangle formed by the lines passing through the origin and the line $12x - 5y + 13 = 0$, in sq. units is

- 1) $3\sqrt{3}$ 2) $2\sqrt{3}$ 3) $\sqrt{3}$ 4) $1/\sqrt{3}$

22. If the sides of a triangle are $ax^2 + 2hxy + by^2 = 0$ and $y = x + c$, then its area is

1) $\frac{c^2 \sqrt{h^2 - ab}}{|a + b + 2h|}$ 2) $\left| \frac{c \sqrt{h^2 - ab}}{a + b + 2h} \right|$

3) $\frac{\sqrt{h^2 - ab}}{|a + b + c|}$ 4) $\frac{\sqrt{h^2 - ab}}{|a + b + 2h|}$

Pair of angular bisectors:

23. The equation of the bisectors of the angle between the two straight lines

$$2x^2 - 3xy + y^2 = 0 \text{ is}$$

- 1) $3x^2 - 2xy + 3y^2 = 0$
2) $x^2 + xy - y^2 = 0$
3) $3x^2 + 2xy + 3y^2 = 0$
4) $3x^2 + 2xy - 3y^2 = 0$

24. If the equation of the pair of bisectors of the angle between the pair of lines

$$3x^2 + xy + by^2 = 0 \text{ is } x^2 - 14xy - y^2 = 0 \text{ then } b =$$

- 1) 4 2) -4 3) 8 4) -8

25. If the lines $x^2 + (2 + k)xy - 4y^2 = 0$ are equally inclined to the coordinate axes, then k =

- 1) -1 2) -2 3) -3 4) -4

26. If the pair of straight lines

$$x^2 - 2pxy - y^2 = 0 \text{ and } x^2 - 2qxy - y^2 = 0 \text{ be such that each pair bisects the angle between the other pair, then}$$

- 1) $pq = -1$ 2) $p = q$
3) $p = -q$ 4) $pq = 1$

27. If one of the lines in the pair of straight lines given by $4x^2 + 6xy + ky^2 = 0$ bisects the angle between the coordinate axes, then $k \in$

(EAM-2011)

- 1) $\{-2, -10\}$ 2) $\{-2, 10\}$ 3) $\{-10, 2\}$ 4) $\{2, 10\}$

Equally inclined with a line:

28. If $x^2 - y^2 = 0$, $lx + 2y = 1$ form an isosceles triangle then $l =$

- 1) 1 2) 2 3) 3 4) 0

29. If the two pairs of lines $2x^2 + 6xy + y^2 = 0$ and $4x^2 + 18xy + by^2 = 0$ are equally inclined, then b =

- 1) 1 2) -1 3) 2 4) -2

Non - homogenous pair of lines:

30. Two lines $9x^2 + y^2 + 6xy - 4 = 0$ are

- 1) parallel and coincident
- 2) coincident only
- 3) parallel but not coincident
- 4) perpendicular

Slopes of pair of lines:

31. If the lines represented by

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ are}$$

written in the form $y = m_1x + c_1$ and

$$y = m_2x + c_2, \text{ then}$$

- 1) $m_1 + m_2 = a/b, m_1m_2 = 2h/b$
- 2) $m_1 + m_2 = -2h/b, m_1m_2 = a/b$
- 3) $m_1 + m_2 = -2h/b, m_1m_2 = -b/a$
- 4) $m_1 + m_2 = 2h/b, m_1m_2 = a/b$

Condition to represent a pair of lines:

32. The value k such that $3x^2 + 11xy + 10y^2 + 7x + 13y + k = 0$ represents a pair of straight lines is

- 1) 1 2) 2 3) 3 4) 4

33. If $kx^2 + 10xy + 3y^2 - 15x - 21y + 18 = 0$ represents a pair of straight lines then k =

- 1) 3 2) 4 3) -3 4) 5

34. If $x^2 + 4xy + 4y^2 + 4x + cy + 3 = 0$ can be written as a product of two linear factors then c =

- 1) 2 2) 3 3) 8 4) 4

35. The condition that the equation

$$ax^2 + by^2 + c(x + y) = 0 \text{ to represents a}$$

pair of straight lines is

- 1) $a + b = 0$ or $c = 0$
- 2) $a + b = 0$ or $c = 0, ab < 0$
- 3) $ab > 0, c = 0$ 4) $a + b \neq 0, c \neq 0$

36. If $x^2 + \alpha y^2 + 2\beta y = a^2$ represents a pair of perpendicular lines, then $\beta =$ (EAM-2014)

- 1) 2a 2) 3a 3) 4a 4) a

Angle between the pair of lines:

37. Angle between the pair of lines

$$2x^2 - 7xy + 3y^2 + 3x + y - 2 = 0$$

- 1) $\pi/2$ 2) $\pi/3$ 3) $\pi/4$ 4) $\pi/6$

38. The equation

$$x^2 - 5xy + py^2 + 3x - 8y + 2 = 0 \text{ represents a pair of straight lines. If } \theta \text{ is the angle between them, then } \sin \theta = \text{ (EAM-2013)}$$

- 1) $\frac{1}{\sqrt{50}}$ 2) $\frac{1}{7}$ 3) $\frac{1}{5}$ 4) $\frac{1}{\sqrt{10}}$

39. The acute angle between the lines

$$(5x - 2y)^2 - 3(2x + 5y)^2 = 0 \text{ is}$$

- 1) $\pi/6$ 2) $\pi/4$ 3) $\pi/3$ 4) $\pi/2$

40. If the angle between the lines represented by

$$2x^2 + 5xy + 3y^2 + 6x + 7y + 4 = 0 \text{ is}$$

$$\tan^{-1}(m) \text{ and } a^2 + b^2 - ab - a - b + 1 \leq 0, \text{ then } 2a + 3b =$$

- 1) $1/m$ 2) m 3) $-m$ 4) m^2

41. If the angle between the pair of lines

$$2x^2 + \lambda xy + 3y^2 + 8x + 14y + 8 = 0 \text{ is } \frac{\pi}{4}, \text{ then the value of } \lambda \text{ is}$$

- 1) ± 6 2) ± 7 3) ± 5 4) ± 1

Pair of parallel and perpendicular lines:

42. The equation to the pair of straight lines passing through (2, 1) and perpendicular to the pair of lines $4xy + 2x + 6y + 3 = 0$ is

- 1) $xy + x + 2y + 2 = 0$ 2) $xy + x + 2y - 2 = 0$
- 3) $xy + x - 2y - 2 = 0$ 4) $xy - x - 2y + 2 = 0$

43. If the equation $2x^2 + 5xy + 3y^2 + 6x + 7y + 4 = 0$ represents a pair of lines, then the equation of pair of lines parallel to them and passing through the point (0,1) is

- 1) $2x^2 + 5xy + 3y^2 - 5x + 6y = 0$
- 2) $2x^2 + 5xy + 3y^2 + 5x - 6y + 3 = 0$
- 3) $2x^2 + 5xy + 3y^2 - 5x - 6y + 3 = 0$
- 4) $2x^2 + 5xy + 3y^2 + 5x + 6y + 3 = 0$

Distance between parallel lines:

44. If the distance between the pair of parallel lines $x^2 + 2xy + y^2 - 8ax - 8ay - 9a^2 = 0$ is $25\sqrt{2}$ then $a =$

1) 1 2) 2 3) 3 4) 5

45. The condition for the equation $x^2 + (\lambda + \mu)xy + \lambda\mu y^2 + x + \mu y = 0$ to represent pair of parallel lines and the distance between them are

1) $\lambda + \mu = 0, \sqrt{1 + \mu^2}$ 2) $\lambda + \mu = 0, \frac{1}{\sqrt{1 + \mu^2}}$

3) $\lambda = \mu, \sqrt{1 + \mu^2}$ 4) $\lambda = \mu, \frac{1}{\sqrt{1 + \mu^2}}$

46. The distance between the parallel lines given by $(x + 7y)^2 + 4\sqrt{2}(x + 7y) - 42 = 0$ is (EAM-2012)

1) $4/5$ 2) $4\sqrt{2}$ 3) 2 4) $10\sqrt{2}$

Product of perpendiculars:

47. The product of perpendicular distances from the origin to the pair of straight lines

$$12x^2 + 25xy + 12y^2 + 10x + 11y + 2 = 0$$

1) $\frac{1}{25}$ 2) $\frac{2}{25}$ 3) $\frac{3}{25}$ 4) $\frac{4}{25}$

Point of intersection of the pair of lines:

48. The point of the intersection of the pair of lines $x^2 + xy + 2y^2 - 3x + 2y + 4 = 0$ is

1) (1, 2) 2) (-1, 2) 3) (-2, 1) 4) (2, -1)

49. The square of the distance of the point of intersection of the lines

$$6x^2 - 5xy - 6y^2 + x + 5y - 1 = 0$$

is

1) $74/169$ 2) $85/169$ 3) $74/185$ 4) $2/13$

50. If the pair of straight lines

$$xy - x - y + 1 = 0$$

and the line $ax + 2y - 3 = 0$ are concurrent then $a =$

1) -1 2) 3 3) 1 4) 0

51. If the lines represented by

$$ax^2 + 4xy + y^2 + 8x + 2y + c = 0$$

intersect on Y-axis, then $(f, c) =$

1) (2, 4) 2) (4, 2) 3) (-2, -4) 4) (-4, -2)

Intercepts of a pair of lines on coordinate axes:

52. The intercept made by the pair of lines $6x^2 - 7xy - 3y^2 - 24x - 3y + 18 = 0$ on the X-axis is

1) 2 2) 4 3) 6 4) 8

53. A: Intercept made by the pair of lines

$$6x^2 - 7xy - 3y^2 - 24x + 3y + 18 = 0$$

on X-axis

B: Intercept made by the pair of lines

$$2x^2 + 4xy - 6y^2 + 3x + y + 1 = 0$$

on Y-axis

C: The distance between the parallel lines

$$x^2 + 2\sqrt{3}xy + 3y^2 - 3x - 3\sqrt{3}y - 4 = 0$$

Arrange A, B, C in descending order

1) A, B, C 2) C, A, B 3) B, C, A 4) C, B, A

Nature of quadrilateral:

54. The figure formed by the four lines

$$3x^2 + 10xy + 3y^2 = 0$$

$$3x^2 + 10xy + 3y^2 - 28x - 28y + 49 = 0$$

1) parallelogram 2) rhombus

3) rectangle 4) square

55. If the adjacent sides of a parallelogram are $2x^2 - 5xy + 3y^2 = 0$ and one diagonal is $x + y + 2 = 0$ then the other diagonal is

1) $9x - 11y = 0$ 2) $9x + 11y = 0$

3) $11x - 9y = 0$ 4) $11x + 9y = 0$

56. The area of the square formed by the lines

$$6x^2 - 5xy - 6y^2 = 0$$

$$6x^2 - 5xy - 6y^2 + x + 5y - 1 = 0$$

in sq. units is

1) $1/\sqrt{3}$ 2) $4/\sqrt{13}$ 3) $\sqrt{13}$ 4) $1/13$

Homogenisation :

57. The equation to the pair of lines joining the origin to the points of intersection of $y = x + 3$ and $2x^2 + 2y^2 = 1$ is

$$1) 12(x^2 + y^2) = (x - y)^2$$

$$2) 6(x^2 + y^2) = (x - y)^2$$

$$3) 18(x^2 + y^2) = (x - y)^2$$

$$4) 2(x^2 + y^2) = (x - y)^2$$

58. A pair of perpendicular straight lines passes through the origin and also through the point of intersection of the curve $x^2 + y^2 = 4$ with $x + y = a$. The set containing the value of 'a' is

(EAM-08,10)

1) $\{-2, 2\}$ 2) $\{-3, 3\}$ 3) $\{-4, 4\}$ 4) $\{-5, 5\}$

59. The angle between the pair of straight lines formed by joining the points of intersection of

$$x^2 + y^2 = 4 \text{ and } y = 3x + c$$

to the origin is a right angle. Then c^2 is equal to (EAM-2007)

- 1) 20 2) 13 3) $1/5$ 4) 5

Types of triangles :

60. The triangle formed by the pair of lines

$$3x^2 + 48xy + 23y^2 = 0 \text{ and the line}$$

$$3x - 2y + 4 = 0 \text{ is}$$

- 1) Equilateral 2) Isosceles
3) Right angled 4) Scalane

LEVEL-I (C.W)-KEY

- | | | | | |
|-------|-------|-------|-------|-------|
| 1) 2 | 2) 3 | 3) 2 | 4) 2 | 5) 1 |
| 6) 3 | 7) 1 | 8) 3 | 9) 3 | 10) 2 |
| 11) 4 | 12) 3 | 13) 3 | 14) 2 | 15) 3 |
| 16) 1 | 17) 4 | 18) 3 | 19) 4 | 20) 1 |
| 21) 4 | 22) 1 | 23) 4 | 24) 2 | 25) 2 |
| 26) 1 | 27) 3 | 28) 4 | 29) 1 | 30) 3 |
| 31) 2 | 32) 4 | 33) 1 | 34) 3 | 35) 2 |
| 36) 4 | 37) 3 | 38) 1 | 39) 3 | 40) 1 |
| 41) 2 | 42) 4 | 43) 3 | 44) 4 | 45) 4 |
| 46) 3 | 47) 2 | 48) 4 | 49) 4 | 50) 3 |
| 51) 1 | 52) 1 | 53) 2 | 54) 2 | 55) 1 |
| 56) 4 | 57) 3 | 58) 1 | 59) 1 | 60) 1 |

LEVEL-I (C.W)-HINTS

- Separate the given equation by resolving into factors
- $h^2 - ab > 0 \Rightarrow 1 - 4a^2 > 0 \Rightarrow a^2 - \frac{1}{4} < 0$
- $\frac{2\sqrt{h^2 - ab}}{|b|}$
- Put $\theta = 45^\circ$ and use $|m_1 - m_2| = \frac{2\sqrt{h^2 - ab}}{|b|}$
- $y^2 - (m_1 + m_2)xy + m_1m_2x^2 = 0$
- $4kh^2 = (k+1)^2 ab$ & $k = 2$
- $y^2 - (m_1 + m_2)xy + m_1m_2x^2 = 0$
- A: $4kh^2 = (k+1)^2 ab$ & $k = 3$

$$R: (m+n)^2 ab = 4h^2 mn$$

$$9. x^2 + 2cxy + y^2 = 0$$

$$\text{Here } m_1 + m_2 = 8m_1m_2, c = -2$$

$$10. \cos \theta = \frac{|a+b|}{\sqrt{(a-b)^2 + 4h^2}}$$

$$11. a+b=0$$

$$12. a+b=0$$

$$13. \sin \theta = \frac{2\sqrt{h^2 - ab}}{\sqrt{(a-b)^2 + 4h^2}}$$

$$14. a(x-x_1)^2 + 2h(x-x_1)(y-y_1) + b(y-y_1)^2 = 0$$

$$15. bx^2 - 2hxy + ay^2 = 0$$

$$16. \frac{|a\alpha^2 + 2h\alpha\beta + b\beta^2|}{\sqrt{(a-b)^2 + 4h^2}}$$

$$17. \frac{|a\alpha^2 + 2h\alpha\beta + b\beta^2|}{\sqrt{(a-b)^2 + 4h^2}}$$

$$18. \frac{n^2\sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$$

$$19. \frac{n^2\sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$$

$$20. \frac{n^2\sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$$

$$21. \frac{c^2}{\sqrt{3}(a^2 + b^2)}$$

$$22. \frac{n^2\sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$$

$$23. h(x^2 - y^2) = (a-b)xy$$

$$24. h(x^2 - y^2) = (a-b)xy$$

$$25. h = 0$$

$$26. \text{Equation of the bisector of the angles between}$$

$$x^2 - 2pxy - y^2 = 0 \text{ is}$$

$$px^2 + 2xy - py^2 = 0 \text{ this same as}$$

$$x^2 - 2qxy - y^2 = 0 \therefore \frac{p}{1} = \frac{1}{-q} = \frac{p}{1} \Rightarrow pq = -1$$

$$27. (a+b)^2 = 4h^2$$

28. $h(l^2 - m^2) = (a - b)lm$
 29. Both the pairs have same pair of angular bisectors
 30. $(3x + y)^2 = 2^2 \Rightarrow 3x + y = 2, 3x + y = -2$
 31. $m_1 + m_2 = -2h/b, m_1 m_2 = a/b$
 32. $\Delta = 0$
 33. $\Delta = 0$
 34. $\Delta = 0$
 35. $\Delta = 0, h^2 > ab$
 36. $a + b = 0$ and $\Delta = 0$

37. $\cos \theta = \frac{|a + b|}{\sqrt{(a - b)^2 + 4h^2}}$

38. $\Delta = 0$ and $\sin \theta = \frac{2\sqrt{h^2 - ab}}{\sqrt{(a - b)^2 + 4h^2}}$

39. $\cos \theta = \frac{|a + b|}{\sqrt{(a - b)^2 + 4h^2}}$

40. Here $\tan \theta = \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right|$

$\therefore \theta = \tan^{-1} \left(\frac{1}{5} \right) = \tan^{-1}(m)$

$\therefore m = \frac{1}{5}$ from the given condition we get
 $a = 1, b = 1$

41. $\cos \theta = \frac{|a + b|}{\sqrt{(a - b)^2 + 4h^2}}$

42. $b(x - x_1)^2 - 2h(x - x_1)(y - y_1) + a(y - y_1)^2 = 0$

43. $a(x - x_1)^2 + 2h(x - x_1)(y - y_1) + b(y - y_1)^2 = 0$

44. $2\sqrt{\frac{g^2 - ac}{a(a + b)}}$

45. $h^2 = ab$ and distance $= 2\sqrt{\frac{g^2 - ac}{a(a + b)}}$

46. distance $= 2\sqrt{\frac{g^2 - ac}{a(a + b)}}$

47. $\frac{|c|}{\sqrt{(a - b)^2 + 4h^2}}$

48. $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$

49. $\frac{f^2 + g^2}{a^2 + h^2}$

50. Point of intersection is $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$ and
 substitute in line

51. $f^2 = bc, hf = bg$

52. $\frac{2\sqrt{g^2 - ac}}{|a|}$

53. $A: \frac{2\sqrt{g^2 - ac}}{|a|}$

$B: \frac{2\sqrt{f^2 - bc}}{|b|}$

$C: 2\sqrt{\frac{g^2 - ac}{a(a + b)}}$

54. $a + b \neq 0, (a - b)fg + h(f^2 - g^2) = 0$

55. $(bl - hm)y = (am - hl)x$

56. $\frac{|c|}{2\sqrt{h^2 - ab}}$

57. $2x^2 + 2y^2 = \left(\frac{y - x}{3} \right)^2$

58. $n^2(a + b) - 2n(lg + mf) + c(l^2 + m^2) = 0$

59. $n^2(a + b) - 2n(lg + mf) + c(l^2 + m^2) = 0$

60. The pair of lines passing through origin and which
 forms an equilateral triangle with the given line is

$(3x - 2y)^2 - 3(2x + 3y)^2 = 0$

$\Rightarrow 3x^2 + 48xy + 23y^2 = 0$



LEVEL - I (H.W)

Homogeneous Pair of lines Separate equations of pair of lines:

1. The separate equations of the lines represented by the equation

$$(2x+3y)^2 - (x-2y)^2 = 0 \text{ are}$$

- 1) $3x+y=0; x+5y=0$ 2) $3x-y=0; x-5y=0$
3) $x+3y=0; 5x-y=0$ 4) $x+y=0; x-5y=0$

Nature of pair of lines:

2. The equation $3x^2 + 10xy - 8y^2 = 0$ represents

- 1) real and distinct lines 2) coincident lines
3) imaginary lines 4) parallel lines

Slopes of pair of lines:

3. If $6x^2 - 5xy + y^2 = 0$ represents a pair of lines then

I: $m_1 + m_2 = 5$ II: $|m_1 - m_2| = 1$

Which of the above statements are correct

- 1) only I 2) only II
3) both I and II 4) neither I nor II

4. If s and p are respectively the sum and the product of the slope of the lines

$$3x^2 - 2xy - 15y^2 = 0, \text{ then } s:p = \text{ (EAM-2011)}$$

- 1) 4:3 2) 2:3 3) 3:5 4) 3:4

5. The difference of the slopes of the lines represented by

$$x^2 (\tan^2 \theta + \cos^2 \theta) + 2xy \tan \theta + y^2 \sin^2 \theta = 0 \text{ is}$$

- 1) 1 2) 2 3) 3 4) 4

6. The combined equation of the lines passing through the origin and having slopes 3 and -2 is

- 1) $6x^2 - xy + y^2 = 0$ 2) $x^2 + xy - 6y^2 = 0$
3) $6x^2 + xy - y^2 = 0$ 4) $x^2 - xy + 6y^2 = 0$

7. If the slopes of the lines represented by $ax^2 + 2hxy + by^2 = 0$ are in the ratio 3 : 2, then

- 1) $25ab = 24h^2$ 2) $8h^2 = 9ab$
3) $16h^2 = 25ab$ 4) $h^2 = ab$

8. If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then c has the value (AIEEE-2004)

- 1) 2 2) -1 3) 1 4) -2

Angle between pair of lines:

9. If the acute angle between the pair of lines

$$2x^2 + 5xy + 3y^2 = 0 \text{ is } \tan^{-1} k \text{ then } k =$$

- 1) $1/5$ 2) 1 3) $7/5$ 4) 7

10. If the angle 2θ is acute, then the acute angle between the pair of straight lines

$$x^2 (\cos \theta - \sin \theta) + 2xy \cos \theta + y^2 (\cos \theta + \sin \theta) = 0 \text{ is}$$

(EAM - 2002)

- 1) 2θ 2) $\frac{\theta}{2}$ 3) $\frac{\theta}{3}$ 4) θ

11. If the pair of lines represented by

$$(x^2 + y^2) \tan^2 \alpha = (x - y \tan \alpha)^2 \text{ are}$$

perpendicular to each other, then $\alpha =$

- 1) $\pi/6$ 2) $\pi/3$ 3) $\pi/8$ 4) $\pi/4$

12. If θ is the acute angle between the lines

$$6x^2 + 11xy + 3y^2 = 0, \text{ then } \tan \theta =$$

- 1) $9/7$ 2) $7/9$ 3) $3/7$ 4) $7/3$

Parallel and perpendicular lines :

13. The equation of the pair of lines passing through (1,2) and parallel to the coordinate axes is

- 1) $xy - 2x - y + 2 = 0$ 2) $xy - x - 2y + 2 = 0$
3) $xy + 2x + y + 2 = 0$ 4) $xy + x + 2y + 2 = 0$

14. The equation of the pair of the lines through (1, -1) and perpendicular to the pair of lines

$$x^2 - xy - 2y^2 = 0 \text{ is}$$

1) $2x^2 - xy + y^2 + 5x + y + 2 = 0$

2) $2x^2 - xy - y^2 - 5x - y + 2 = 0$

3) $x^2 - xy + 2y^2 - 5x - y - 2 = 0$

4) $2x^2 - xy - y^2 + 5x + y - 2 = 0$

Product of perpendiculars:

15. The product of perpendiculars from (0,1) to the pair of lines $2x^2 - 5xy + y^2 = 0$ is

- 1) $\frac{1}{2}$ 2) $\frac{1}{\sqrt{26}}$ 3) $\frac{1}{4}$ 4) $\frac{1}{\sqrt{13}}$

16. If the product of the perpendicular distances from (1, k) to the pair of lines $x^2 - 4xy + y^2 = 0$ is $3/2$ units, then $k =$

- 1) 4 2) 5 3) 6 4) 8

Area of the Triangle:

17. The area of the triangle formed by the lines $x^2 + 4xy + y^2 = 0$, $x+y=1$ is

1) $\sqrt{3}$ 2) 2 3) 1 4) $\sqrt{3}/2$

18. The area (in square units) of the triangle formed by $x + y + 1 = 0$ and the pair of straight lines $x^2 - 3xy + 2y^2 = 0$ is: (EAM-2009)

1) $7/12$ 2) $5/12$ 3) $1/12$ 4) $1/6$

19. Match the following :

List - I (sides of the triangle) List - II (Area of the triangle)

A. $x^2 - 4xy + y^2 = 0$ and $x + y = 1$ 1) $\frac{1}{2} sq.u$

B. $xy = 0$ and $x + y = 1$ 2) $\frac{1}{2\sqrt{3}} sq.u$

C. $x^2 + 4xy + y^2 = 0$ and $x + y = 1$ 3) $\frac{\sqrt{3}}{2} sq.u$
4) $\frac{1}{\sqrt{3}} sq.u$

A	B	C	A	B	C
1) 2	1	4	2) 2	1	3
3) 1	2	4	4) 1	2	3

20. The area (in square units) of the triangle

formed by the lines $x^2 - 3xy + y^2 = 0$ and $x + y + 1 = 0$ (EAM-2014)

1) $5\sqrt{2}$ 2) $\frac{1}{2\sqrt{5}}$ 3) $\frac{2}{\sqrt{3}}$ 4) $\frac{\sqrt{3}}{2}$

21. The area of the equilateral triangle

formed by the lines $x^2 - 3y^2 = 0$ and the line $x - 3a = 0$, in sq. units is

1) $\sqrt{3}a^2$ 2) $2\sqrt{3}a^2$ 3) $3\sqrt{3}a^2$ 4) $4\sqrt{3}a^2$

22. If a, h, b are in A.P. then the triangle area formed by the pair of lines

$ax^2 + 2hx + by^2 = 0$ and the line $x - y = -2$ in square units is

1) $\left \frac{a+b}{a-b} \right $	2) $\left \frac{a^2+b^2}{a-b} \right $
3) $\left \frac{a-b}{a+b} \right $	4) $\left \frac{a^2+b^2}{a+b} \right $

Pair of angular bisectors:

23. The equation of the bisectors of the angle between the two straight lines

$$x^2 - xy - 6y^2 = 0 \text{ is}$$

1) $x^2 + 14xy - y^2 = 0$ 2) $x^2 + 14xy + y^2 = 0$

3) $x^2 - 14xy - y^2 = 0$ 4) $x^2 - 14xy + y^2 = 0$

24. The pair of straight lines $h(x^2 - y^2) + pxy = 0$ bisects the angle between the pair of lines

$$ax^2 + 2hxy + by^2 = 0 \text{ then the value of p is}$$

1) $a - b$ 2) $b - a$ 3) $a + b$ 4) $-a - b$

25. If one of the lines $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$, then m is (AIEEE-2007)

1) $-\frac{1}{2}$ 2) -2 3) 1 4) 2

26. If one of the lines in the pair of straight lines given by $x^2 + (2 + k)xy - 4y^2 = 0$ bisects the angle between the coordinate axes, then $k \in$

1) $\{-1, -5\}$ 2) $\{-1, 5\}$ 3) $\{1, -5\}$ 4) $\{1, 5\}$

27. If the pairs of lines $3x^2 - 2pxy - 3y^2 = 0$ and $5x^2 - 2qxy - 5y^2 = 0$ are such that each pair bisects the angle between the other pair, then pq equals to

1) -1 2) -7 3) -9 4) -15

28. The equation of the pair of bisectors of the angles between the pair of lines $x^2 + 2axy - y^2 = 0$ is $x^2 + 2bxy - y^2 = 0$. Then

1) $ab = 1$ 2) $ab + 1 = 0$

3) $ab = 2$ 4) $ab + 2 = 0$

Equally inclined with a line:

29. If $2x + 3y = 7$ makes equal angles with $9x^2 + 12xy + ky^2 = 0$ then k =

1) 3 2) 7 3) 14 4) 21

30. The pair of lines $2x^2 + 3xy + 5y^2 = 0$; $4x^2 + 21xy + 25y^2 = 0$ are

1) mutually perpendicular

2) equally inclined to the axes

3) equally inclined to each other

4) Each pair bisects the angle between the other

Non homogeneous pair of lines slopes:

31. $2x^2 + 5xy + 3y^2 + 6x + 7y + 4 = 0$ represents two lines $y = m_1x + c_1$ and $y = m_2x + c_2$ then $m_1 + m_2$ and $m_1 m_2$ are
- 1) $-5/3, -2/3$ 2) $-5/3, 2/3$
 3) $5/3, -2/3$ 4) $5/3, 2/3$

Condition for pair of lines

32. If $4xy + 2x + 2fy + 3 = 0$ represents a pair of lines then $f =$
 1) 2 2) 3 3) 4 4) 5
33. If the equation $2xy + gx + fy + 5 = 0$ represents a pair of lines, then $fg =$
 1) 1 2) 10 3) 5 4) 0
34. The value of λ such that $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$ represents a pair of straight lines is (EAM-2008)
 1) 1 2) -1 3) 2 4) -2
35. The value of λ with $|\lambda| < 16$ such that $2x^2 - 10xy + 12y^2 + 5x + \lambda y - 3 = 0$ represents a pair of straight lines is (EAM-2009)
 1) -10 2) -9 3) 10 4) 9
36. $x^2 + k_1y^2 + 2k_2y = a^2$ represents a pair of perpendicular lines if
 1) $k_1 = 1, k_2 = a$ 2) $k_1 = 1, k_2 = -a$
 3) $k_1 = -1, k_2 = -a$ 4) $k_1 = -1, k_2 = a^2$

Angle between pair of lines:

37. The angle between the pair of lines $2x^2 + 5xy + 2y^2 + 3x + 3y + 1 = 0$ is
 1) $\cos^{-1}\left(\frac{4}{5}\right)$ 2) $\tan^{-1}\left(\frac{4}{5}\right)$
 3) 0 4) $\pi/2$
38. The angle between the pair of lines $2(x+2)^2 + 3(x+2)(y-2) - 2(y-2)^2 = 0$ is
 1) $\pi/4$ 2) $\pi/3$ 3) $\pi/6$ 4) $\pi/2$
39. The equation $x^2 - 3xy + \lambda y^2 + 3x - 5y + 2 = 0$ when λ is a real number, represents a pair of straight lines. If θ is the angle between these lines, then $\operatorname{Cosec}^2 \theta =$
 1) 3 2) 9 3) 10 4) 100

Parallel and perpendicular lines:

40. The equation to the pair of lines through the origin perpendicular to the pair of lines $2x^2 + 5xy + 2y^2 + 10x + 5y = 0$ is
 1) $2x^2 + 5xy + 2y^2 = 0$
 2) $2x^2 - 5xy + 2y^2 = 0$
 3) $2x^2 - 5xy - 2y^2 = 0$
 4) $x^2 - 5xy + y^2 = 0$
41. The equation of the pair of lines through the point (a, b) and parallel to the coordinate axes is
 1) $(x-b)(y-a) = 0$ 2) $(x-a)(y+b) = 0$
 3) $(x-a)(y-b) = 0$ 4) $(x+a)(y-b) = 0$

Distance between parallel lines :

42. If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of parallel lines then

$$\sqrt{\frac{g^2 - ac}{f^2 - bc}} = \quad \quad \quad \text{(EAM-2011)}$$

1) $\frac{a}{b}$ 2) $\sqrt{\frac{a}{b}}$ 3) $\sqrt{\frac{b}{a}}$ 4) $\frac{b}{a}$

43. The distance between the parallel lines $9x^2 - 6xy + y^2 + 18x - 6y + 8 = 0$ is
 1) $\frac{1}{\sqrt{10}}$ 2) $\frac{2}{\sqrt{10}}$ 3) $\frac{4}{\sqrt{10}}$ 4) $\sqrt{10}$
44. The distance between the parallel lines $x^2 + 2\sqrt{2}xy + 2y^2 + 4x + 4\sqrt{2}y + 1 = 0$ is
 1) 2 2) $2\sqrt{2}$ 3) $4\sqrt{2}$ 4) 8
45. The distance between the two lines represented by $8x^2 - 24xy + 18y^2 - 6x + 9y - 5 = 0$ is (EAM-2010)
 1) 0 2) $\frac{3}{4\sqrt{13}}$ 3) $\frac{6}{\sqrt{13}}$ 4) $\frac{7}{2\sqrt{13}}$

Product of perpendiculars:

46. The product of the perpendicular distances from the point (-2, 3) to the lines $x^2 - y^2 + 2x + 1 = 0$ is
 1) 3 2) 4 3) 5 4) 6

Point of intersection of pair of lines:

47. The point of intersection of the straight lines represented by

$$6x^2 + xy - 40y^2 - 35x - 83y + 11 = 0$$

- 1) (3, 1) 2) (3, -1) 3) (-3, 1) 4) (-3, -1)

48. If $3x^2 - 11xy + 10y^2 - 7x + 13y + k = 0$ denotes a pair of straight lines, then the point of intersection of the lines is (EAM-2010)

- 1) (1, 3) 2) (3, 1) 3) (-3, 1) 4) (1, -3)

49. If the equation

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines, then the square of the distance of their point of intersection from the origin is (EAM-2013)

1) $\frac{c(a+b) - af^2 - bg^2}{ab - h^2}$

2) $\frac{c(a+b) + f^2 + g^2}{ab - h^2}$

3) $\frac{c(a+b) - f^2 - g^2}{ab - h^2}$

4) $\frac{c(a+b) - f^2 - g^2}{(ab - h^2)^2}$

50. A: The product of the perpendiculars from (2, -1) to the lines $2x^2 + 6xy + y^2 = 0$

B: The product of perpendiculars from (1, 1) to the pair of lines

$$3x^2 + 10xy + 3y^2 - 28x - 28y + 49 = 0$$

C: The product of perpendiculars from (0, 0) to pair of lines

$$2x^2 - 13xy - 7y^2 + x + 23y - 6 = 0$$

D: The square of the distance from (0, 0) to the point of intersection of

$$x^2 - y^2 - x - 3y - 2 = 0$$

Arrange A, B, C, D in ascending order

- 1) D, A, B, C 2) B, C, A, D
3) A, B, C, D 4) C, A, B, D

51. If the lines

$$x^2 + 2xy - 35y^2 - 4x + 44y - 12 = 0 \text{ and}$$

$5x + \lambda y - 8 = 0$ are concurrent, then the value of λ is (EAM-2007)

- 1) 0 2) 1 3) -1 4) 2

52. If the pair of lines

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

intersect on the y-axis, then (AIEEE-2002)

1) $2fgh = bg^2 + ch^2$ 2) $bg^2 \neq ch^2$

3) $abc = 2fgh$ 4) $2fgh = af^2 + ch^2$

Intercepts made by pair of lines:

53. The intercept made by the pair of lines $2x^2 + xy - 8y^2 - 2x + 7y + 1 = 0$ on the Y-axis is

- 1) 6 2) 9/8 3) 1/5 4) 5

Nature of Quadrilateral:

54. The figure formed by the pairs of lines $2x^2 + 3xy - 2y^2 = 0$ and $2x^2 + 3xy - 2y^2 - 5x + 15y - 25 = 0$ is a

- 1) parallelogram 2) rhombus
3) rectangle 4) square

55. The pairs of straight lines $x^2 - 3xy + 2y^2 = 0$

and $x^2 - 3xy + 2y^2 + x - 2 = 0$ form a :

(EAM-2009)

- 1) square but not rhombus
2) rhombus 3) parallelogram
4) rectangle but not a square

56. A diagonal of the rectangle formed by the lines

$$x^2 - 7x + 6 = 0 \text{ and } y^2 - 14y + 40 = 0 \text{ is}$$

- 1) $5x + 6y = 0$ 2) $5x - 6y = 0$
3) $6x - 5y + 14 = 0$ 4) $6x - 5y - 14 = 0$

57. The length of the side of the square formed by the lines $2x^2 + 3xy - 2y^2 = 0$ and $2x^2 + 3xy - 2y^2 + 3x + y + 1 = 0$ is

- 1) $\frac{1}{\sqrt{3}}$ 2) $\frac{1}{\sqrt{5}}$ 3) $\frac{1}{\sqrt{7}}$ 4) $\frac{1}{\sqrt{10}}$

Homogenisation:

58. The angle between the lines joining the origin to the points of intersection of the lines

$$\sqrt{3}x + y = 2 \text{ and the curve } x^2 + y^2 = 4 \text{ is}$$

- 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

LEVEL-I (H.W)-KEY

- | | | | | |
|-------|-------|-------|-------|-------|
| 1) 1 | 2) 1 | 3) 3 | 4) 2 | 5) 2 |
| 6) 3 | 7) 1 | 8) 1 | 9) 1 | 10) 4 |
| 11) 4 | 12) 2 | 13) 1 | 14) 2 | 15) 2 |
| 16) 2 | 17) 4 | 18) 3 | 19) 2 | 20) 2 |
| 21) 3 | 22) 3 | 23) 1 | 24) 2 | 25) 3 |
| 26) 3 | 27) 4 | 28) 2 | 29) 3 | 30) 3 |
| 31) 2 | 32) 2 | 33) 2 | 34) 3 | 35) 2 |
| 36) 3 | 37) 1 | 38) 4 | 39) 3 | 40) 2 |
| 41) 3 | 42) 2 | 43) 2 | 44) 1 | 45) 4 |
| 46) 2 | 47) 2 | 48) 2 | 49) 3 | 50) 4 |
| 51) 4 | 52) 1 | 53) 2 | 54) 4 | 55) 3 |
| 56) 3 | 57) 2 | 58) 3 | | |

LEVEL-I (H.W)-HINTS

- $a^2 - b^2 = 0 \Rightarrow (a+b)(a-b) = 0$
- $h^2 - ab > 0$
- I: $m_1 + m_2 = -\frac{2h}{b}$
II: $|m_1 - m_2| = \frac{2\sqrt{h^2 - ab}}{|b|}$
- $s = m_1 + m_2 = \frac{-2h}{b}$ and $p = m_1 m_2 = \frac{a}{b}$
- Put $\theta = 45^\circ$ and use $|m_1 - m_2| = \frac{2\sqrt{h^2 - ab}}{|b|}$
- $y^2 - (m_1 + m_2)xy + m_1 m_2 x^2 = 0$
- $l : m = 3 : 2$ and $\frac{(l+m)^2}{4lm} = \frac{h^2}{ab}$
- $m_1 + m_2 = 4m_1 m_2 \Rightarrow \frac{-2h}{b} = \frac{4a}{b}$
 $\Rightarrow c = 2$
- $\tan \theta = \frac{2\sqrt{h^2 - ab}}{|a+b|}$
- $\cos \alpha = \frac{|a+b|}{\sqrt{(a-b)^2 + 4h^2}}$
- $a + b = 0$
- $\tan \theta = \frac{2\sqrt{h^2 - ab}}{|a+b|}$

13. $a(x-x_1)^2 + 2h(x-x_1)(y-y_1) + b(y-y_1)^2 = 0$

14. $b(x-x_1)^2 - 2h(x-x_1)(y-y_1) + a(y-y_1)^2 = 0$

15. $\frac{|a\alpha^2 + 2h\alpha\beta + b\beta^2|}{\sqrt{(a-b)^2 + 4h^2}}$

16. $\frac{|a\alpha^2 + 2h\alpha\beta + b\beta^2|}{\sqrt{(a-b)^2 + 4h^2}}$

17. $\frac{n^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$

18. $\frac{n^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$

19. $\frac{n^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$

20. $\frac{n^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$

21. $\frac{c^2}{\sqrt{3}(a^2 + b^2)}$

22. $2h = a + b$ and apply $\frac{n^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$

23. $h(x^2 - y^2) = (a-b)xy$

24. Compare the equation of pair of angular bisectors of second pair with first pair

 25. One of the line represented by $ax^2 + 2hxy + by^2 = 0$ are bisectors of angles between the coordinate axes (i.e, $x=0, y=0$)

$$\text{then } (a+b)^2 = 4h^2 \Rightarrow (m-m)^2 = (1-m^2)^2$$

$$\Rightarrow 1-m^2 = 0 \Rightarrow m = 1$$

26. $(a+b)^2 = 4h^2$

27. The equation of bisectors of

$$3x^2 - 2pxy - 3y^2 = 0 \text{ is } \frac{x^2 - y^2}{3 - (-3)} = \frac{xy}{-p}$$

$$\text{But this is same as } 5x^2 - 2qxy - 5y^2 = 0$$

$$\therefore \frac{6}{p} = -\frac{2q}{5}$$

28. $h(x^2 - y^2) = (a - b)xy$
29. $h(l^2 - m^2) = (a - b)lm$
30. Both the pairs have same pair of angular bisectors
31. $m_1 + m_2 = \frac{-2h}{b}, m_1 m_2 = \frac{a}{b}$
32. $\Delta = 0$
33. $\Delta = 0$
34. $\Delta = 0$
35. $\Delta = 0$
36. $a + b = 0, \Delta = 0$
37. $\cos \theta = \frac{|a+b|}{\sqrt{(a-b)^2 + 4h^2}}$
38. $a + b = 0$
39. $\sin \theta = \frac{2\sqrt{h^2 - ab}}{\sqrt{(a-b)^2 + 4h^2}}$
40. $bx^2 - 2hxy + ay^2 = 0$
41. Combined equation of the lines $x = a$ and $y = b$
42. $2\sqrt{\frac{g^2 - ac}{a(a+b)}} = 2\sqrt{\frac{f^2 - bc}{b(a+b)}}$
43. $2\sqrt{\frac{g^2 - ac}{a(a+b)}}$
44. $2\sqrt{\frac{g^2 - ac}{a(a+b)}}$
45. distance $= 2\sqrt{\frac{g^2 - ac}{a(a+b)}}$
46. $\frac{|a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c|}{\sqrt{(a-b)^2 + 4h^2}}$
47. $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0$
48. $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0$
49. $\Delta = 0$ and P.I $= \left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$

50. $A: \frac{|a\alpha^2 + 2h\alpha\beta + b\beta^2|}{\sqrt{(a-b)^2 + 4h^2}}$
- $B: \frac{|a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c|}{\sqrt{(a-b)^2 + 4h^2}}$
- $C: \frac{|c|}{\sqrt{(a-b)^2 + 4h^2}}$
- $D: \frac{f^2 + g^2}{a^2 + h^2}$
51. P is the point of intersection of $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0$ and p lies on the line
52. Let $f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ (1)
- Point of intersection of lines $\frac{\partial f(x, y)}{\partial x} = 0$
- $\Rightarrow 2ax + 2hy + 2g = 0$
- Since, it intersects on y-axis i.e., $x = 0$
- $\Rightarrow y = \frac{-g}{h}$
- Thus putting this value in Eq. (1) we get
- $bg^2 + ch^2 = 2fgh$
53. $\frac{2\sqrt{f^2 - bc}}{|b|}$
54. $a+b=0, (a-b)fg+h(f^2 - g^2)=0$
55. $a+b \neq 0, (a-b)fg+h(f^2 - g^2) \neq 0$
56. vertices of the rectangle are A(1,4) B(6,4) C(6,10) and D(1,10) find $\overline{AC}$ or $\overline{BD}$
57. $a^2 = \frac{|c|}{2\sqrt{h^2 - ab}}$ then find a
58. Homogenising
- $x^2 + y^2 - \frac{(\sqrt{3}x + y)^2}{4} = 0$
- $\Rightarrow 2x^2 + 2\sqrt{3}xy = 0$

$$\cos \theta = \cos \left(\frac{\pi}{3} \right) \Rightarrow \theta = \frac{\pi}{3}$$

**LEVEL - II (C.W)****Angle between pair of lines:**

- If the pair of lines $2x^2 + 3xy + y^2 = 0$ makes angles θ_1 and θ_2 with X-axis then $\tan(\theta_1 - \theta_2) =$
1) 1 2) $1/2$ 3) $1/3$ 4) $1/4$
- The triangle formed by $x + 3y = 1$ and $9x^2 - 12xy + ky^2 = 0$ is right angled triangle and $k \neq -9$. Then $k =$
1) 3 2) 5 3) 7 4) 1

Equally inclined:

- If the equation $2x^2 - 5xy + 2y^2 = 0$ represents two sides of an isosceles triangle then the equation of the third side passing through the point (3,3) is
1) $x+y=3$ 2) $x-y=0$
3) $2x-y=3$ 4) $x+y-6=0$

Condition of perpendicular pair:

- If $px^2 - y^2 + 3x + 11y + q = 0$ represents a pair of perpendicular lines then $(p, q) =$
1) $(-1, -14)$ 2) $(-1, 28)$
3) $(1, -28)$ 4) $(-1, -28)$
- The pair of lines represented by $3ax^2 + 5xy + (a^2 - 2)y^2 = 0$ are perpendicular to each other for
1) Two values of a 2) for all values of a
3) for one value of a 4) for no value of a

- If the equation $x^2 + py^2 + y = a^2$ represents a pair of perpendicular lines, then the point of intersection of the lines is
1) $(1, a)$ 2) $(1, -a)$ 3) $(0, a)$ 4) $(0, 2a)$

Common line to pair of lines:

- If the pair of lines $3x^2 - 5xy + py^2 = 0$ and $6x^2 - xy - 5y^2 = 0$ have one line in common, then $p =$
1) $2, \frac{25}{4}$ 2) $-2, \frac{25}{4}$
3) $-2, \frac{-25}{4}$ 4) $2, \frac{-25}{4}$

- If one of the lines represents by $3x^2 - 4xy + y^2 = 0$ is perpendicular to one of the line $2x^2 - 5xy + ky^2 = 0$ then $k =$

- 1) -3, $13/9$ 2) 3, $-13/9$
3) -3, $-13/9$ 4) -7, -33

- The condition that one of the pair of lines $ax^2 + 2hxy + by^2 = 0$ be coincident with one line of the pair $3x^2 + 12xy + 2y^2 = 0$ and the remaining lines are at right angles, then $h(a-b) =$

- 1) $a+b$ 2) ab 3) $2ab$ 4) a/b

Pair of angular bisectors:

- The line $y = 3x$ bisects the angle between the lines $ax^2 + 2axy + y^2 = 0$ if $a =$
1) 3 2) 11 3) $\frac{3}{11}$ 4) $\frac{11}{3}$

Centres related with triangles:

- The centroid of the triangle formed by the pair of straight lines $12x^2 - 20xy + 7y^2 = 0$ and the line $2x - 3y + 4 = 0$ is (EAM-2006)
1) $\left(-\frac{7}{3}, \frac{7}{3}\right)$ 2) $\left(-\frac{8}{3}, \frac{8}{3}\right)$
3) $\left(\frac{8}{3}, \frac{8}{3}\right)$ 4) $\left(\frac{4}{3}, \frac{4}{3}\right)$
- If $2x^2 - 5xy + 2y^2 = 0$ represents two sides of a triangle whose centroid is (1, 1) then the equation of the third side is
1) $x+y-3=0$ 2) $x-y-3=0$
3) $x+y+3=0$ 4) $x-y+3=0$
- The orthocentre of the triangle formed by the lines $x^2 - 3y^2 = 0$ and the line $x=a$ is
1) $\left(\frac{a}{3}, 0\right)$ 2) $\left(\frac{2a}{3}, 0\right)$
3) $(a, 0)$ 4) $\left(\frac{4a}{3}, 0\right)$
- If $x^2 + 4xy + y^2 = 0$ represents two sides of $\triangle OAB$ and the orthocentre is $(-1, -1)$, then the third side is
1) $x+y=2$ 2) $x+y=1$
3) $x+y+1=0$ 4) $x+y=3$

15. The circumcentre of the triangle formed by the lines $x^2 - y^2 = 0$ and $y - 5 = 0$ is

1) (5, 0) 2) (0, 5) 3) (0, 0) 4) (5, 5)

16. The orthocentre of the triangle formed by the lines $x + 3y - 10 = 0$ and

$6x^2 + xy - y^2 = 0$ is (EAM-2001)

1) (1, 3) 2) (3, 1) 3) (-1, 3) 4) (1, -3)

P.I. of pair of lines

17. The distance from the point of intersection of the lines $x^2 - y^2 - 2x + 2y = 0$ to the point of intersection of the lines

$2x^2 - 5xy + 2y^2 + x + y - 1 = 0$ is

1) 1 2) 0 3) 2 4) 4

Homogenisation:

18. If $\frac{x}{a} + \frac{y}{b} = 1$ intersects

$5x^2 + 5y^2 + 5bx + 5ay - 9ab = 0$ at P and Q,

$\angle POQ = \pi/2$ then the relation between a and b is

1) $a = b$ 2) $a = 2b$ or $b = 2a$

3) $a = 3b$ or $b = 3a$ 4) $a + b = 5$

19. If the pair of lines which joins the origin to the point of intersection of

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$,

$a_1x^2 + 2h_1xy + b_1y^2 + 2g_1x + 2f_1y + c_1 = 0$ are at right angles then

1) $\frac{g}{g_1} = \frac{a_1 + b_1}{a + b}$ 2) $\frac{g}{g_1} = \frac{a + b}{a_1 + b_1}$

3) $\frac{h}{h_1} = \frac{a + b}{a_1 + b_1}$ 4) $\frac{h}{h_1} = \frac{a_1 + b_1}{a + b}$

20. The angle between the lines joining the origin to the point of intersection of $lx + my = 1$ and $x^2 + y^2 = a^2$ is

1) $\frac{\pi}{2}$ 2) $\frac{\pi}{4}$

3) $\cos^{-1}\left(\frac{1}{a\sqrt{l^2 + m^2}}\right)$ 4) $2\cos^{-1}\left(\frac{1}{a\sqrt{l^2 + m^2}}\right)$

21. The combined equation of the pair of lines passing through origin which are at a distance 4 units from the point (5, 6) is

1) $9x^2 + 60xy - 20y^2 = 0$

2) $9x^2 - 60xy + 20y^2 = 0$

3) $20x^2 + 60xy - 9y^2 = 0$

4) $20x^2 - 60xy + 9y^2 = 0$

22. Perpendiculars AL, AM are drawn from any point A on the x-axis to the pair of lines $2x^2 - 5xy - 3y^2 = 0$ the angle made by LM with +ve direction of x-axis is

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{2}$

Cubic equation:

23. Two of the lines represented by

$ax^3 + 3bx^2y + 3cxy^2 + dy^3 = 0$ will be

perpendicular if

1) $a^2 + ac + db + d^2 = 0$

2) $a^2 + 3(ac + bd) + d^2 = 0$

3) $a^2 - 3(ac + bd) + d^2 = 0$

4) $a^2 - ac - bd + d^2 = 0$

24. The line $x + y = 1$ meets the lines represented by the equation

$y^3 - xy^2 - 14x^2y + 24x^3 = 0$ at the points

A, B, C. If O is the point of intersection of the lines represented by the given equation then

$OA^2 + OB^2 + OC^2 =$

1) $\frac{22}{9}$ 2) $\frac{85}{72}$ 3) $\frac{181}{72}$ 4) $\frac{221}{72}$

Biquadratic equations:

25. If two lines represented by

$x^4 + x^3y + cx^2y^2 - xy^3 + y^4 = 0$ bisect the angle between the other two, then the value of 'c' is

1) 0 2) -1 3) 1 4) -6

26. If a and b positive numbers ($a < b$), then the range of values of k for which a real λ be found such that equation

$$ax^2 + 2\lambda xy + by^2 + 2k(x+y+1) = 0$$

represents a pair of straight lines is

- 1) $a < k^2 < b$ 2) $a \leq k^2 \leq b$
 3) $k^2 \leq a$ or $k^2 \geq b$ 4) $k \leq 2a$ or $k \geq 2b$

LEVEL-II (C.W) - KEY

- 1) 3 2) 1 3) 4 4) 3 5) 1
 6) 3 7) 4 8) 4 9) 2 10) 3
 11) 3 12) 1 13) 2 14) 2 15) 2
 16) 1 17) 2 18) 2 19) 2 20) 4
 21) 4 22) 3 23) 2 24) 4 25) 4
 26) 4

LEVEL-II (C.W) - HINTS

- $\tan(\theta_1 - \theta_2) = \tan \theta = \frac{2\sqrt{h^2 - ab}}{|a+b|}$
- One of the lines of $ax^2 + 2hxy + by^2 = 0$ is perpendicular to $lx + my + n = 0$ then $al^2 + 2hlm + bm^2 = 0$
- One of the angular bisectors of the given pair of lines is parallel to the third side and passing through (3,3)
- $a + b = 0$ and $\Delta = 0$
- These lines are $\perp^r$, if $a + b = 0$
 $\Rightarrow 3a + (a^2 - 2) = 0$
 $\Rightarrow a^2 + 3a - 2 = 0$
 This is quadratic equation in 'a' and $B^2 - 4AC > 0$
 $\therefore$ the roots of the above equation are real and distinct
 $\therefore$ the lines are $\perp^r$ to each other for two values of 'a'
- $a + b = 0$, $\Delta = 0$ and P.I. $= \left(\frac{-g}{a}, \frac{-f}{b} \right)$
- $(a_1b_2 - a_2b_1)^2 + 4(h_1a_2 - h_2a_1)(h_1b_2 - h_2b_1) = 0$
- $(a_1a_2 - b_1b_2)^2 + 4(h_1a_2 + h_2a_1)(h_1b_2 + h_2b_1) = 0$
- $h_1 \left(\frac{1}{a_1} - \frac{1}{b_1} \right) = h_2 \left(\frac{1}{a_2} - \frac{1}{b_2} \right)$
- Equation of the bisectors of the angles between the lines $ax^2 + 2axy + y^2 = 0$ is

$$\frac{x^2 - y^2}{a - 1} = \frac{xy}{a} \text{ which is satisfied by } y = 3x$$

$$\text{If } \frac{1-9}{a-1} = \frac{3}{a} \Rightarrow a = \frac{3}{11}$$

11. Multiplying the option with $3/2$ and put in the given line

$$12. S_1 = \frac{3}{2} S_{11}$$

$$13. x + \sqrt{3}y = 0; x - \sqrt{3}y = 0; x = a$$

vertices are $(0, 0), (a, a/\sqrt{3}), (a, -a/\sqrt{3})$

In an equilateral triangle, centroid = orthocentre

$$14. (a+b)(cx+dy) = ad^2 - 2hcd + bc^2$$

15. In a right angle triangle mid point of hypotenuse = circumcentre

vertices are $A(0, 0) B(-5, 5) C(5, 5)$

clearly ABC is a right angled triangle

$\therefore$ circumcentre = midpoint of BC

16. Take the perpendicular line to the given line $x + 3y - 10 = 0$ and verify the options

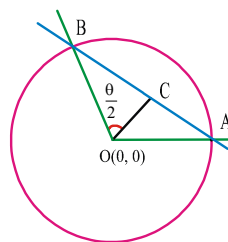
$$17. \frac{\partial f}{\partial x} = 0 \text{ and } \frac{\partial f}{\partial y} = 0$$

$$18. n^2(a+b) - 2n(lg+mf) + c(l^2+m^2) = 0$$

$$19. g_1(ax^2 + 2hxy + by^2 + 2gx) = 0$$

$$-g(a_1x^2 + 2h_1xy + b_1y^2 + 2g_1x) = 0$$

$$20. \Delta OBC, \cos\left(\frac{\theta}{2}\right) = \frac{OC}{OB} = \frac{1}{\sqrt{l^2 + m^2}}$$



$$21. (\beta x - \alpha y)^2 = d^2(x^2 + y^2) \text{ where}$$

$$(\alpha, \beta) = (5, 6)$$

$$22. \theta = \tan^{-1} \left(\frac{b-a}{2h} \right)$$

$$23. m_1m_2m_3 = -\frac{a}{d} \text{ and put } y = \frac{a}{d}x \text{ in given equation}$$

24. The given cubic can be written as $(y-2x)(y-3x)(y+4x)=0$
 $\therefore$ The three lines given by this equation are $y=2x$, $y=3x$ and $y=-4x$, they intersect at $O(0,0)$ and meet the line $x+y=1$ at the points

$$A\left(\frac{1}{3}, \frac{2}{3}\right), B\left(\frac{1}{4}, \frac{3}{4}\right), C\left(\frac{-1}{3}, \frac{4}{3}\right)$$

$$\therefore OA^2 + OB^2 + OC^2 = \frac{5}{9} + \frac{10}{16} + \frac{17}{9} = \frac{221}{72}$$

25. Product of the slopes = 1 and a pair of lines represents the bisectors of the angles between the other two, the product of slopes of each pair is -1. Let the equation of one pair be

$$ax^2 + 2hxy + by^2 = 0, \frac{x^2 - y^2}{2a} = \frac{xy}{h}$$

$$x^4 + x^3y + cx^2y^2 - xy^3 + y^4$$

$$= (ax^2 + 2hxy - ay^2)(hx^2 - 2axy - hy^2)$$

$$ah = 1 \text{ and } c = -6. ah = -6$$

26. We have $h = \lambda, g = k, c = 2k; f = k$, apply $\Delta = 0$ will get quadratic in λ for real $\lambda \Rightarrow b^2 - 4ac \geq 0$



LEVEL-II (H.W)

Angle between pair of lines:

1. The triangle formed by the pair of lines $x^2 - 4y^2 = 0$ and the line $x-a=0$ is always
- Equilateral
 - Isosceles
 - Right angled
 - Scalene

Equally inclined:

2. The lines $33y^2 - 136xy + 135x^2 = 0$ are equally inclined to
- $x+2y+7=0$
 - $2x+y-7=0$
 - $x+2y-7=0$
 - $x+y=1$

Condition for pair of lines:

3. If the equation $6x^2 + 5xy + by^2 + 9x + 20y + c = 0$ represents a pair of perpendicular lines, then $b - c =$
- 6
 - 3
 - 2
 - 0
4. If $ax^2 + 6xy + by^2 - 10x + 10y - 6 = 0$ represents a pair of perpendicular straight lines, then $|a|$ is equal to
- 2
 - 4
 - 1
 - 3

5. If the equation $\lambda x^2 - 5xy + 6y^2 + x - 3y = 0$ represents a pair of straight lines then their point of intersection is (EAM-2000)

- (-3, -1)
- (-1, -3)
- (3, 1)
- (1, 3)

Common line to pair of lines:

6. If the two pairs of lines $3x^2 - 5xy + 2y^2 = 0$ and $6x^2 - xy + ky^2 = 0$ have one line in common, then $k^2 + 7k - 10 =$

- 0
- 20
- 1
- 2

7. If one of the lines represented by $2x^2 + 2hxy + 3y^2 = 0$ be perpendicular to one of the lines given by

$$3x^2 + 2h^1xy + 2y^2 = 0, \text{ then}$$

- $h - h^1 = 0$
- $h + h^1 = 0$
- $3h = 2h^1$
- $2h = 3h^1$

8. If one of the lines given by

$$6x^2 - xy + 4cy^2 = 0 \text{ is } 3x + 4y = 0, \text{ then c equals}$$

- 1
- 1
- 3
- 3

Centres related with triangles:

9. Assertion (A) : If two sides of a triangle are represented by $x^2 - 3xy + 2y^2 = 0$

and centroid is $\left(\frac{2}{3}, 0\right)$, then the third side is

$$2x - 3y - 2 = 0$$

Reason (R): If two sides of a triangle are represented by $ax^2 + 2hxy + by^2 = 0$ and

centroid is $\left(\frac{2x_1}{3}, \frac{2y_1}{3}\right)$ then the third side is

$$x(ax_1 + hy_1) + y(hx_1 + by_1) = ax_1^2 + 2hx_1y_1 + by_1^2$$

- Both A and R are true and R is the correct explanation of A
- Both A and R are true and R is not correct explanation of A
- A is true but R is false
- A is false but R is true

10. The orthocentre of the triangle formed by the lines $2x+y=2$ and $2x^2 + 3xy - 2y^2=0$ is

- (4/3, -2/3)
- (1/2, 1)
- (0,0)
- (1, 1)

11. Assertion A: If two sides of a triangle represented by $2x^2 + 4xy - y^2 = 0$ and orthocentre is $(1,1)$ then the third side is $x + y + 3 = 0$

Reason R: If two sides of a triangle represented by $ax^2 + 2hxy + by^2 = 0$ and orthocentre is (c,d) then the third side is

$$(a+b)(cx+dy) = ad^2 - 2hcd + bc^2$$

- 1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true and R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true
12. The circumcentre of the triangle formed by the lines $2x^2 - 3xy - 2y^2 = 0$ and $3x - y = 10$ is
1) $(2,1)$ 2) $(1,-2)$ 3) $(3,-6)$ 4) $(3,-1)$
13. The coordinates of the orthocentre of the triangle formed by the lines

$$2x^2 - 3xy + y^2 = 0 \text{ and } x+y=1 \text{ are is}$$

- 1) $(1,1)$ 2) $(1/2, 1/2)$
3) $(1/3, 1/3)$ 4) $(1/4, 1/4)$

Area of the triangle:

14. The area of the triangle formed by the two rays whose combined equation is $y = |x|$ and the line $x + 2y = 2$.

- 1) 4sq units 2) $\frac{4}{3}$ sq units
3) $\frac{8}{3}$ sq units 4) $\frac{16}{3}$ sq units

Homogenisation:

15. The curve $x^2 + y^2 + 2gx + 2fy + c = 0$ intercepts on the line $lx + my = 1$, a length which subtends a right angle at

$$\text{the origin, then } \frac{lg+mf+1}{l^2+m^2} =$$

- 1) $\frac{c}{2}$ 2) $\frac{-c}{2}$ 3) $\frac{2}{c}$ 4) $\frac{-2}{c}$

16. The lines joining the origin to the points of intersection of $x^2 + y^2 + 2gx + c = 0$ and $x^2 + y^2 + 2fy - c = 0$ are at right angles is

- 1) $g^2 + f^2 = c$ 2) $g^2 - f^2 = 0$
3) $g^2 - f^2 = 2c$ 4) $g^2 + f^2 = c^2$

17. The line $4y - 3x + 48 = 0$ cuts the curve $y^2 = 64x$ in A and B. If AB subtends an angle θ at the origin, then $\tan \theta =$

- 1) $\frac{20}{9}$ 2) $\frac{10}{9}$ 3) $\frac{5}{9}$ 4) $\frac{40}{9}$

Cubic Equation:

18. If the equation

$$ax^3 + 3bx^2y + 3cxy^2 + dy^3 = 0 \quad (a, b, c, d \neq 0)$$

represents three coincident lines, then

- 1) $a = c$ 2) $b = d$
3) $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$ 4) $ac = bd$

19. If the line $y = \sqrt{3}x$ cuts the curve

$$x^3 + y^3 + 3xy + 5x^2 + 3y^2 + 4x + 5y - 1 = 0 \text{ at the points A, B, C, then } OA \cdot OB \cdot OC \text{ is}$$

- 1) $\frac{4}{13}(3\sqrt{3}-1)$ 2) $3\sqrt{3}+1$
3) $\frac{2}{\sqrt{3}}+7$ 4) $\frac{4}{13}(3\sqrt{3}+1)$

Biquadratic Equation:

20. The equation

$$a(x^4 + y^4) - 4bxy(x^2 - y^2) + 6cx^2y^2 = 0$$

represents two pairs of lines at right angles. The two pairs will coincide if

- 1) $b^2 = a + 3c$ 2) $a^2 = b^2 - 3ac$
3) $a^2 + b^2 = 3ac$ 4) $2b^2 = a^2 + 3ac$

LEVEL-II (H.W)-KEY

- 1) 2 2) 2 3) 4 4) 2 5) 1
6) 2 7) 2 8) 4 9) 1 10) 3
11) 1 12) 4 13) 2 14) 2 15) 2 16) 3
17) 1 18) 3 19) 1 20) 4

LEVEL-II (H.W)-HINTS

1. The lines represented by $x^2 - 4y^2 = 0$ are $x + 2y = 0$ (1) $x - 2y = 0$ (2)
The given line equation is $x - a = 0$...(3)

i.e., angle between (1) and (3) = angle between (1) and (2).

2. One line verify with $h(x^2 - y^2) = (a - b)xy$
3. $a + b = 0$ and $\Delta = 0$
4. $a + b = 0$ and $\Delta = 0$
5. $\Delta = 0, \frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$
6. $(a_1b_2 - a_2b_1)^2 + 4(h_1a_2 - h_2a_1)(h_1b_2 - h_2b_1) = 0$
7. Let the lines be $y = mx$ & $y = \frac{-1}{m}x$
then $3m^2 + 2hm + 2 = 0$
 $3m^2 - 2h'm + 2 = 0$ eliminating 'm'
8. $\Rightarrow 6 - \frac{y}{x} + 4c \frac{y^2}{x^2} = 0$
 $\frac{y}{x} = \frac{-3}{4}$
 $\Rightarrow 6 + \frac{3}{4} + 4c \cdot \frac{9}{16} = 0$
 $\Rightarrow c = -3$
9. A : $D = \frac{3}{2} \cdot G = (1, 0)$ satisfies the line
R : $D = \frac{3}{2} \cdot G = (x_1, y_1) \therefore$ the equation of third side is $S_1 = S_{11}$
10. $a + b = 0$, orthocentre is point of intersection of pair of lines
11. Orthocentre is (c, d) then the third side is
 $(a + b)(cx + dy) = ad^2 - 2hcd + bc^2$
12. $a + b = 0$, it is right angle triangle.
Circumcentre lies on $3x - y = 10$
13. (kl, km) is the orthocentre of the triangle formed by the lines $ax^2 + 2hxy + by^2 = 0$
14. Given lines are $x^2 - y^2 = 0$ and
 $x + 2y - 2 = 0$

Area of the triangle =

$$\frac{n^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|} = \frac{4}{3} \text{ sq.units}$$

15. $n^2(a + b) - 2n(lg + mf) + c(l^2 + m^2) = 0$
16. Subtracting the given equations, we get
 $gx - fy + c = 0$. Apply
 $n^2(a + b) - 2n(lg + mf) + c(l^2 + m^2) = 0$
the first equation and straight line
17. Homogenisation and apply $\tan \theta = \frac{2\sqrt{h^2 - ab}}{|a + b|}$
18. Let $ax^3 + 3bx^2y + 3cx^2y + 3cx^2 + dy^3 = 0$
represent three coincident lines; say $y = mx$
 $\Rightarrow m = -\frac{a}{b} = -\frac{b}{c} = -\frac{c}{d} \Rightarrow \frac{a}{b} = \frac{b}{c} = \frac{c}{d}$
19. $\tan \theta = \sqrt{3}$ Line is $\frac{x-0}{1} = \frac{y-0}{\sqrt{3}} = r$ any point
on the line is $\left(\frac{r}{2}, \frac{\sqrt{3}}{2}r\right)$. Where r is distance
from (0, 0) substituting in the curve
 $r^3 \left(\frac{1+3\sqrt{3}}{8}\right) + r^2(\dots) + r(\dots) - 1 = 0$ this is cubic in 'r'
 $\therefore OA \cdot OB \cdot OC = \frac{1}{(1+3\sqrt{3})}(8) = \frac{4}{13}(3\sqrt{3}-1)$
20. Given equation
 $\equiv (ax^2 + pxy - ay^2)(x^2 + qxy - y^2)$
comparing x^3y and x^2y^2 on both sides
 $p + aq = -4b \rightarrow (1)$
 $-2a + pq = 6c \rightarrow (2)$ again if two pairs coincide
 $\frac{p}{a} = q \rightarrow (3)$
from (1) and (3), $q = -\frac{2b}{a}$ and $p = -2b$
 $\therefore (2)$ becomes $-2a + \frac{4b^2}{a} = 6c$
